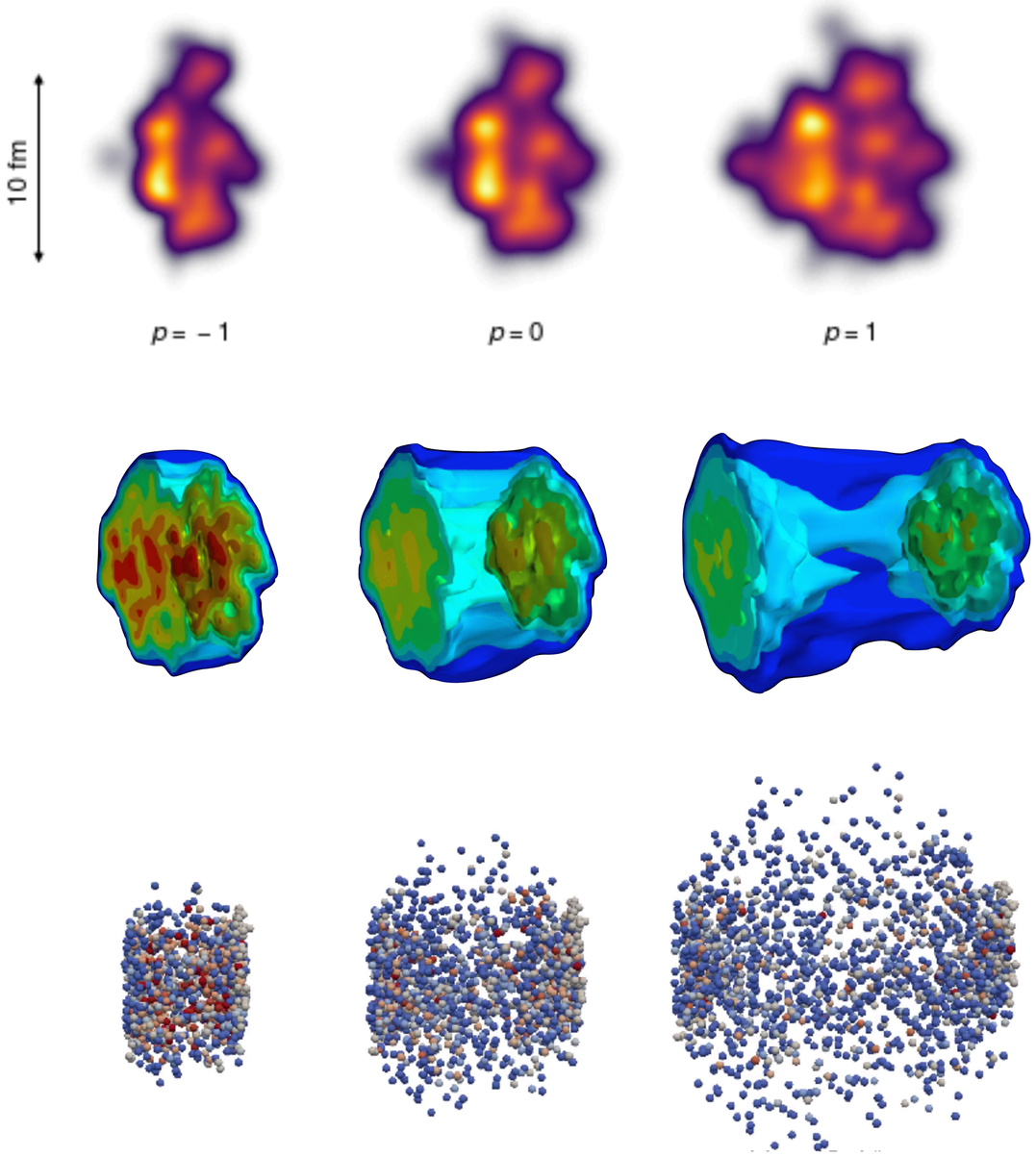



# Heavy Ion Collisions: A speed run.

**Dr. Oscar Garcia-Montero**

Bielefeld Universtität

SS 2023



# Emerging Complexity in Heavy Ion Collisions

**Dr. Oscar Garcia-Montero**

Bielefeld Universtität

SS 2023

# "Contents"

Why Heavy-Ion collisions?

What do we know?

How do we know it?

# Contents

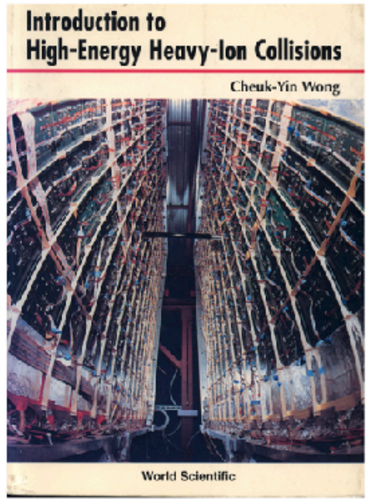

- Why Heavy-Ion collisions? -Introduction
- Basics of Heavy-Ion collisions: Tools, variables and Observables.
- Thermodynamics of the Quark Gluon Plasma
- Space-time Evolution and Modelling of Heavy-Ion Collisions.
    - Introduction to Kinetic Theory
    - Introduction to Kolmogorov's turbulence
    - Introduction to Hydrodynamics
    - Hadronization Models
- Polarization in HICs: Chiral Phenomena.
- Hard Scattering, Jets and Jet Quenching
- Photons and electron pairs: penetrating probes.
- Correlations and Femtoscopy

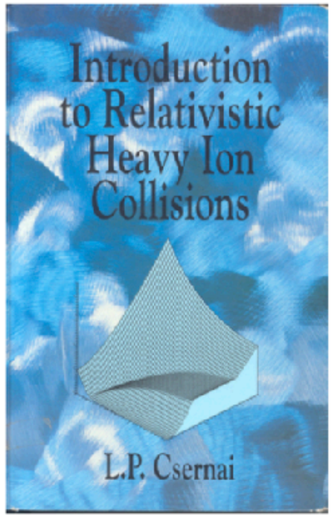
# Audience

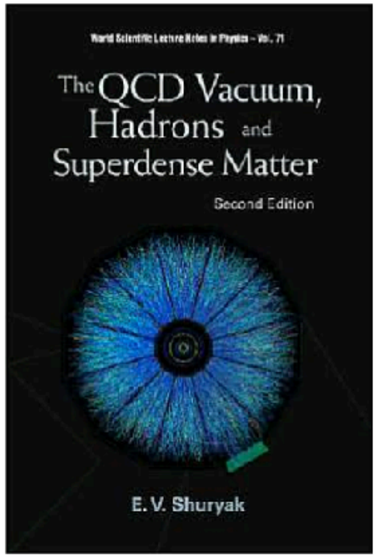
- Bachelor/Master students
  - Deepen knowledge about nuclear and particle physics
  - Learn relativistic kinematics, thermodynamics, basics of QCD …
  - Obtain overview of ultra-relativistic heavy-ion physics
  - Obtain/apply programming skills as part of solving homework assignments (python…)
- PhD students
  - Update on developments in areas besides own research topic
- Attendance and passing
  - Present three exercises and one short (10 min) presentation

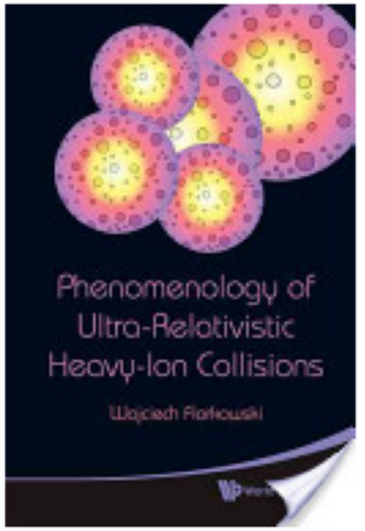
# References


- Phenomenology of Ultra-Relativistic Heavy-Ion Collisions by Wojciech Florkowski.
- Relativistic Hydrodynamics by Luciano Rezzolla and Olindo Zanotti.
- Wong, Introduction to High-Energy Heavy-Ion Collisions,World Scientific, 1994
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- Shuryak, The QCD vacuum, hadrons, and superdense matter, World Scientific, 2004

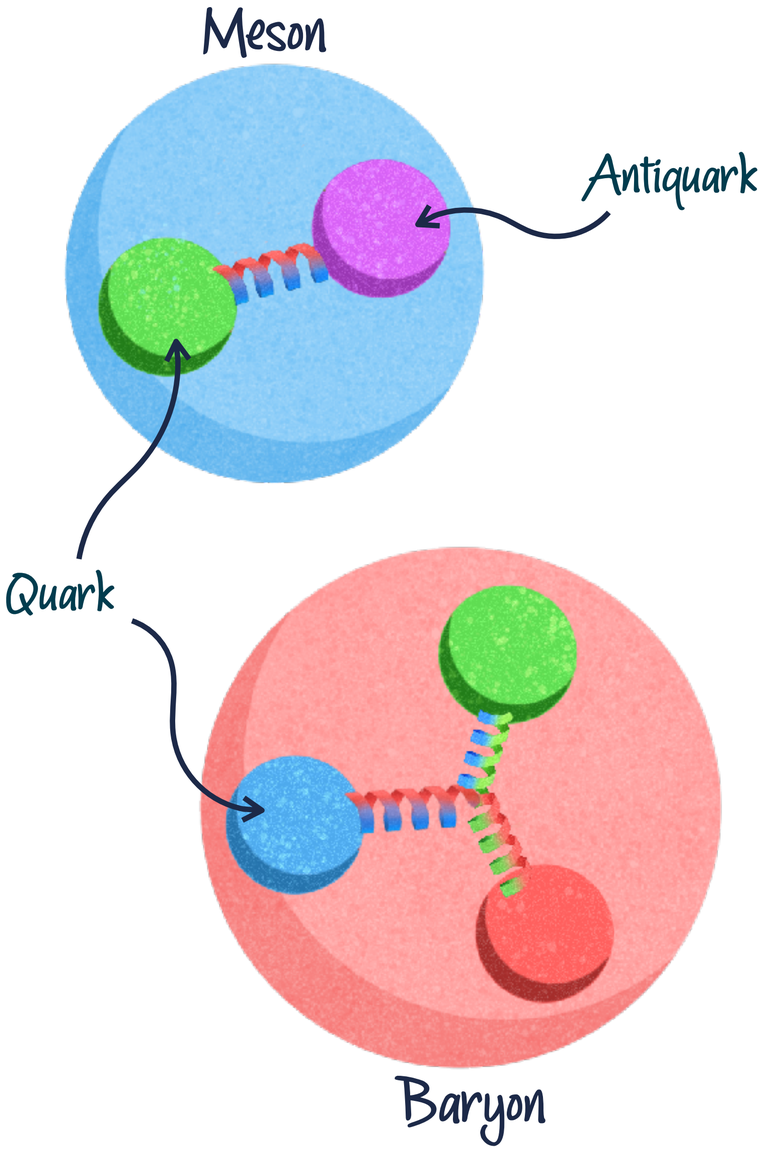



# QCD

## QUANTUM CHROMODYNAMICS




# Remarkable properties of Nuclear Matter

- QCD is **the** theory of strong nuclear interactions
- Hadrons are made of Quarks, (Up, Down, Strange, Charm, Top, Bottom) and gluons

# RUTHERFORD

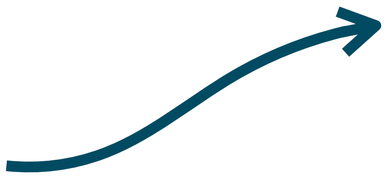


(1920)

RUTHERFORD

(1920)



HEISENBERG

ISOSPIN




$$I_3 = \pm \frac{1}{2}$$

## RUTHERFORD (1920)




## HEISENBERG ISOSPIN




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## PIONS!

An isospin triplet




$$I_3 = (-1,0,1)$$

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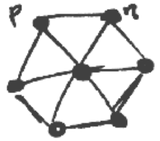
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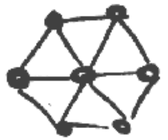
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Gell-Mann's
Eightfold way

SU(3)
symmetric!

**B**





**OCTETS!**

**M**

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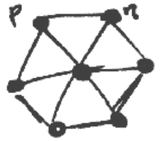
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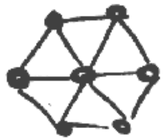
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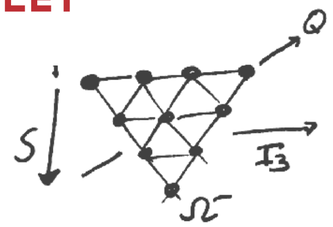
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**DECUPLET**

Gell-Mann
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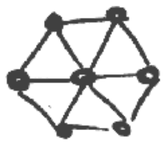
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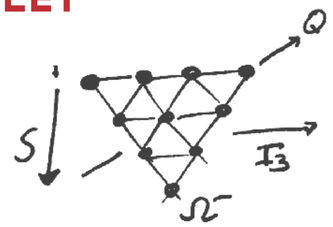
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(1920)

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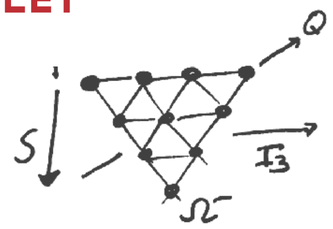
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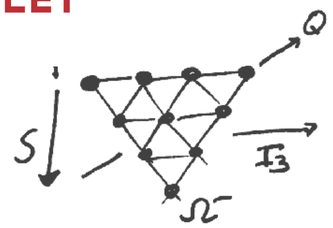
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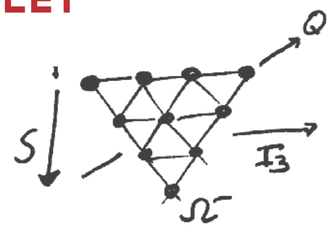
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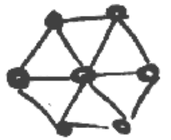
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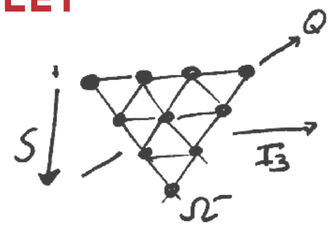
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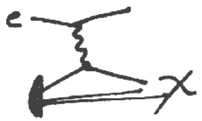
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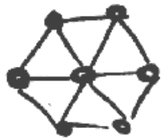
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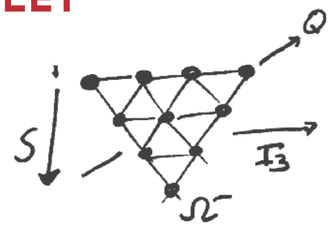
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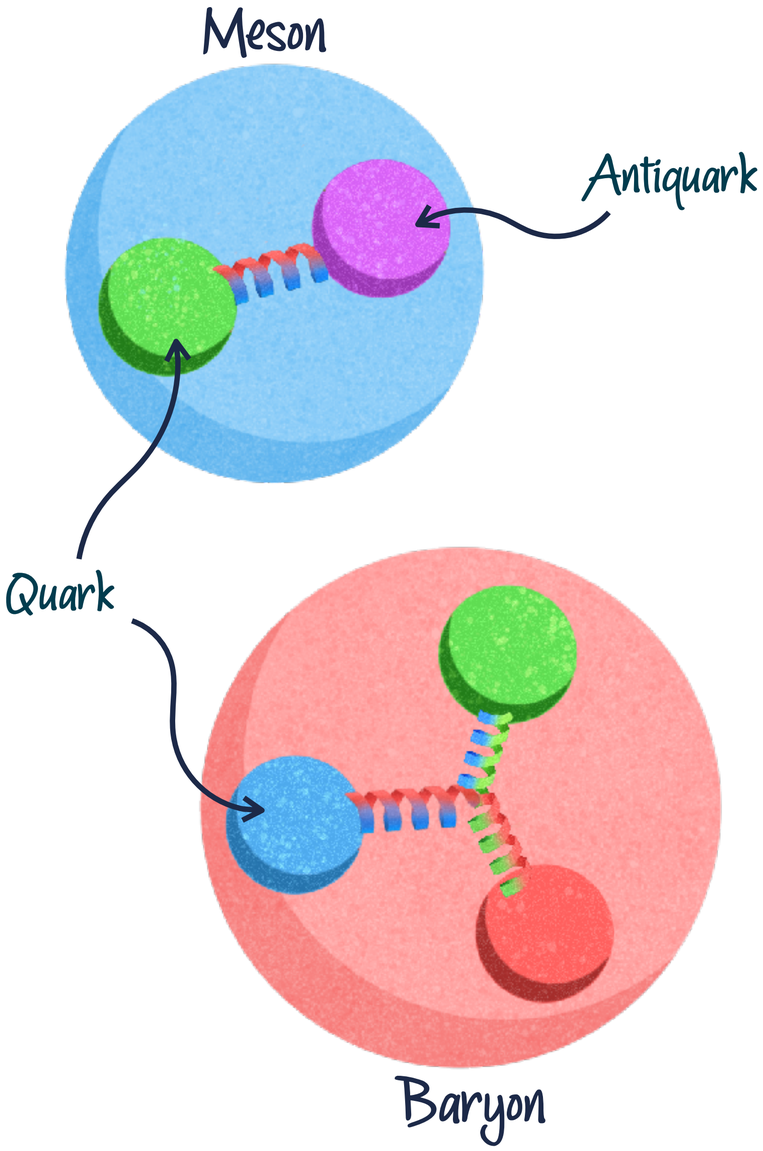
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QCD!

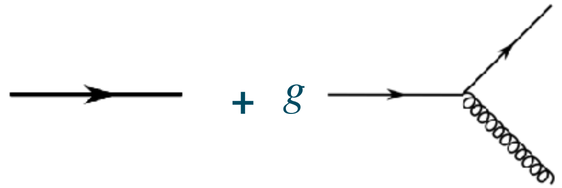



# Remarkable properties of Nuclear Matter

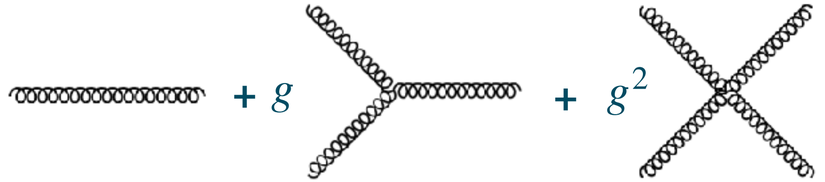
- QCD is **the** theory of strong nuclear interactions
- Hadrons are made of Quarks, (Up, Down, Strange, Charm, Top, Bottom) and gluons
- SU(3) symmetry gives interaction (for both species)

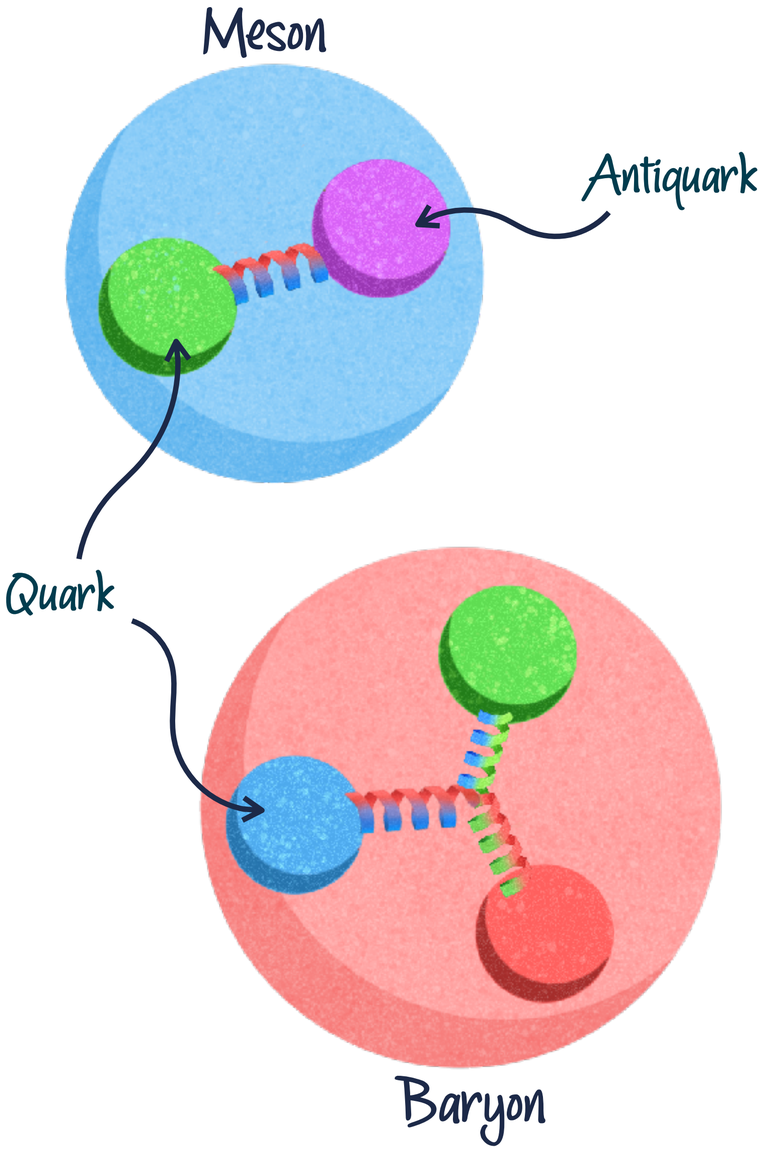
$$\mathscr{L} = \sum_f \bar{\psi}_f \left( i\gamma^\mu D_\mu - m \right) \psi_f - \frac{1}{4} G_a^{\mu\nu} G_{\mu\nu,a}$$

$$\bar{\psi}_f \left( i\gamma^\mu D_\mu - m \right) \psi_f \propto$$



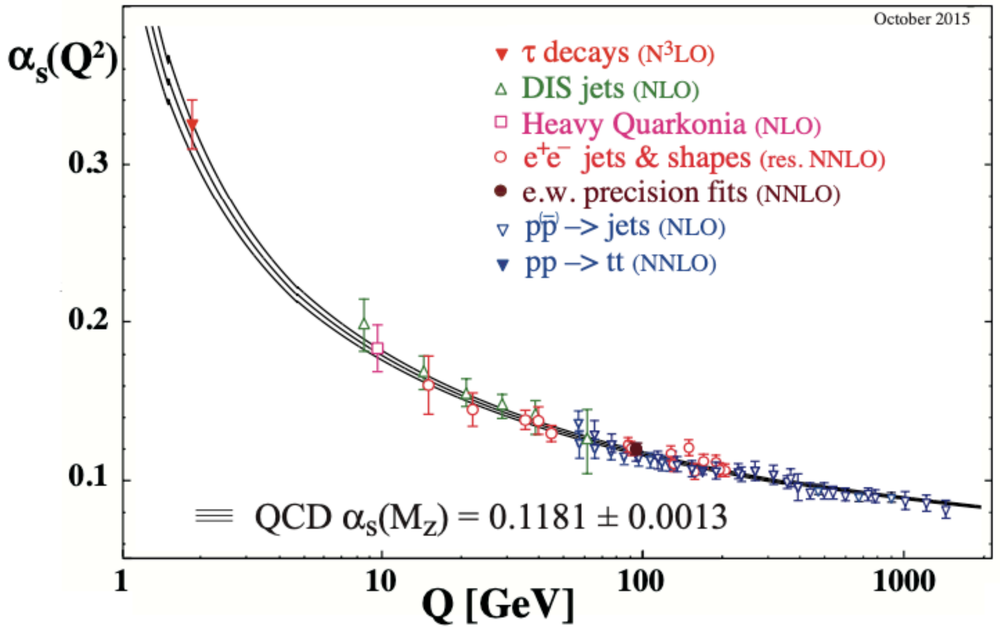

$$\frac{1}{4} G_a^{\mu\nu} G_{\mu\nu,a} \propto$$






# Remarkable properties of Nuclear Matter

- At low energies quarks become trapped (Confinement)
- At high energies quarks become free (Asymptotic Freedom)



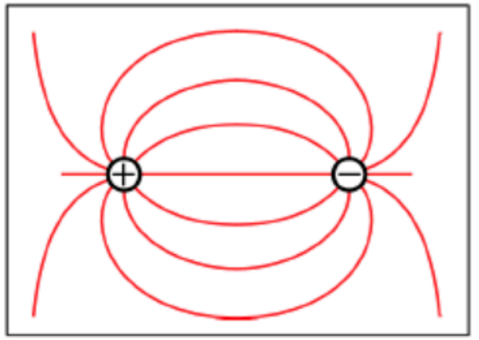

- High sustained energies (like in the Early Universe), one could get an almost free gas of quarks and gluons.

# QED vs QCD

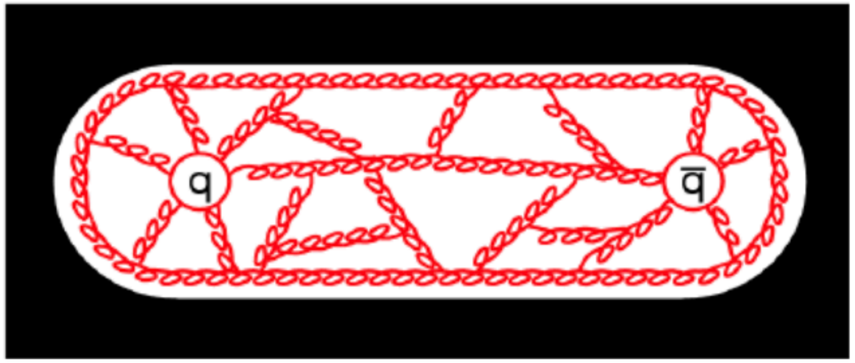
Quarks carry electric charge and color charge (R,G,B) interact strongly by exchange of colored gluons (8 different gluons from 3 colors and 3 anticolors).

QCD is a non- Abelian field theory of Yang-Mills type (1973 Fritzsch, Gell-Mann, Wess).

## QED



## QCD

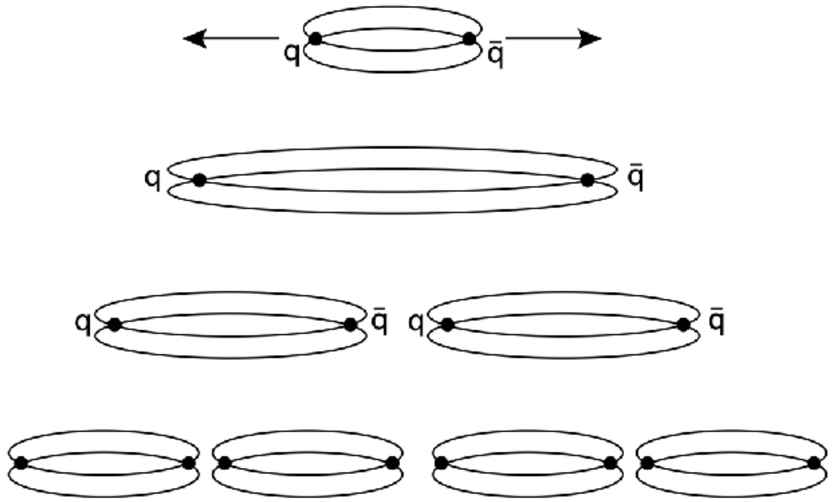


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Quarks are confined in hadrons, trying to pull them apart finally leads to the production of new hadrons



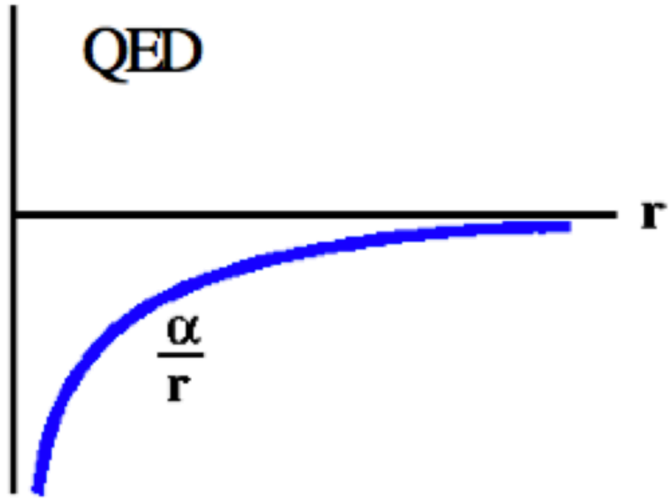

**The “string breaking picture” of QCD**

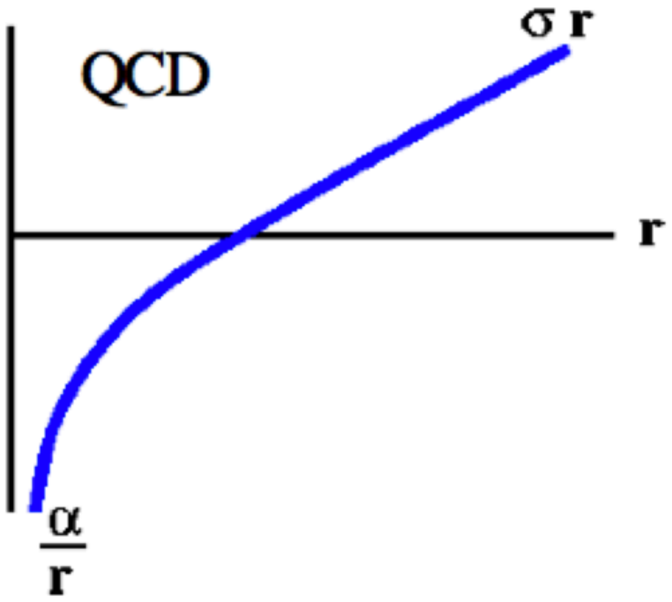
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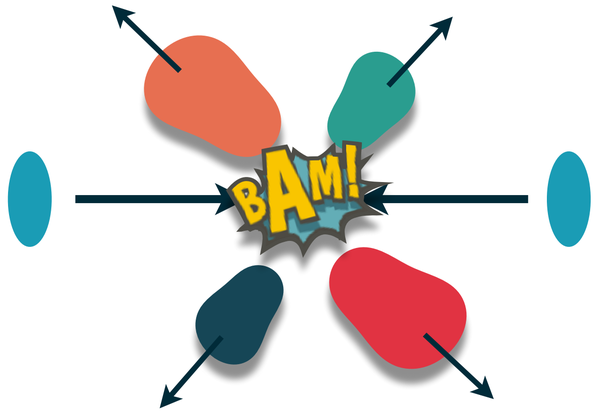
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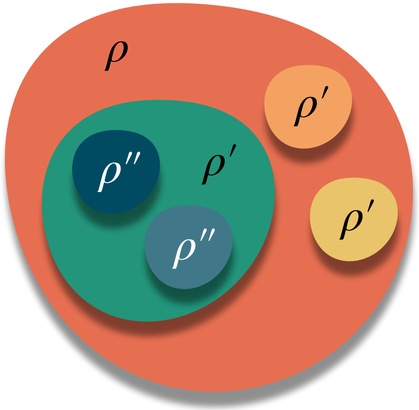



# Limits of the Hadron Gas

## The Bootstrap Model




The bootstrap model was used by Hagedorn to describe the data of elementary collisions, after Fermi's isotropic hadronization idea failed




- 50s, 60s: Number of hadronic states was getting out of hand
  - Idea: Spectral function is self-similar

$$\rho(m) = \delta(m - m_0) + \sum_N \frac{V_0^{N-1}}{N!} \int \prod_{i=1}^{N} dm_i \rho(m_i)\, d^3p_i\, \delta^4\left(\sum_j p_j - p\right)$$

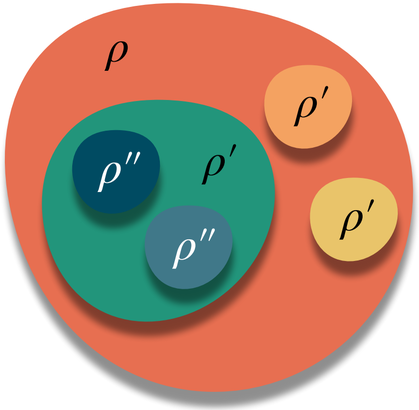
- If one solves for this type of equation: $\rho(m) \approx m^a\, \mathrm{e}^{bm}$
- What is REALLY interesting is the effect on thermodynamics:

$$\epsilon(T) \sim VT^{7/2} \int_m^\infty dm\, m^{5/2-a} \mathrm{e}^{m(b-1/T)} \qquad C_V(T) \sim VT^{3/2} \int_m^\infty dm\, m^{7/2-a} \mathrm{e}^{m(b-1/T)}$$

- Divergence at $T \equiv T_H \sim 0.15\,\mathrm{GeV}$
- Now we know that this corresponds to a change in the degrees of freedom.

# Limits of the Hadron Gas

## The Bootstrap Model




- 50s, 60s: Number of hadronic states was getting out of hand
  - Idea: Spectral function is self-similar

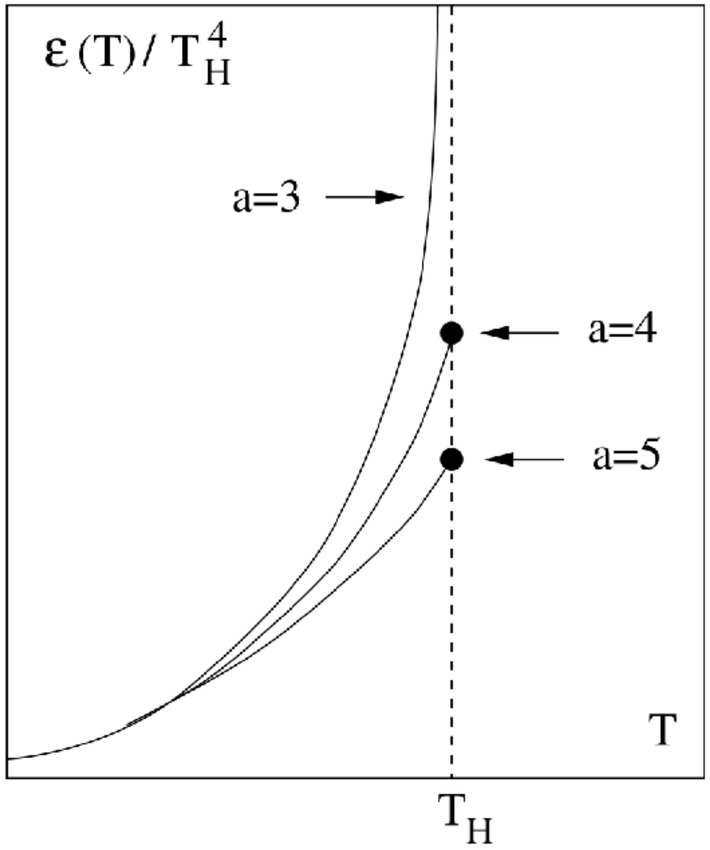
$$\rho(m) = \delta(m - m_0) + \sum_N \frac{V_0^{N-1}}{N!} \int \prod_{i=1}^{N} dm_i \rho(m_i)\, d^3p_i\, \delta^4\left(\sum_j p_j - p\right)$$

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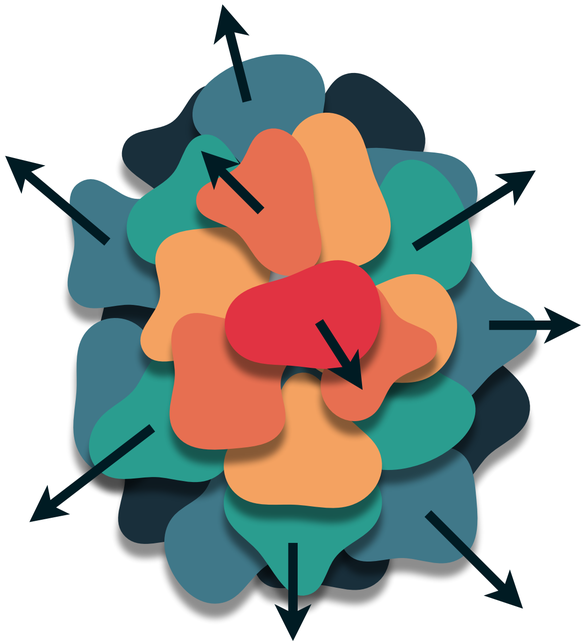
H. Satz, Extreme States of Matter in Strong Interaction Physics, Springer, 2012



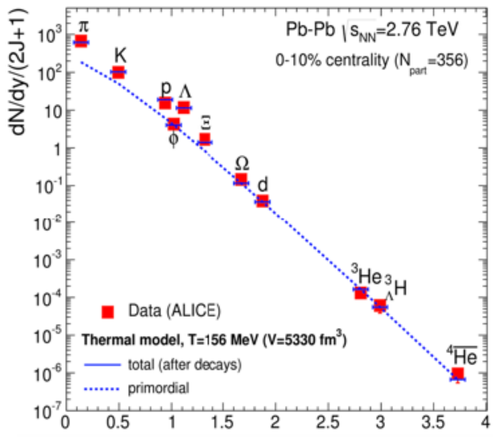
# A Thermal Spectrum

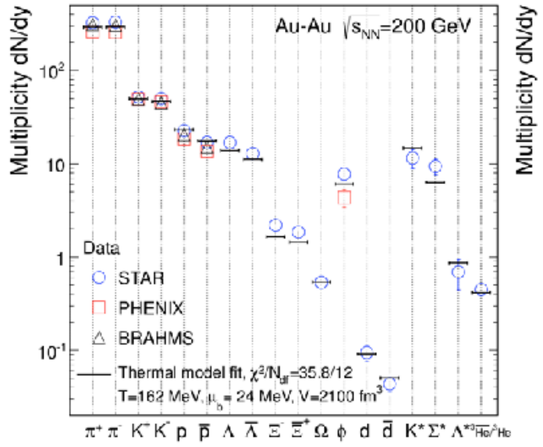
## The Fireball Model

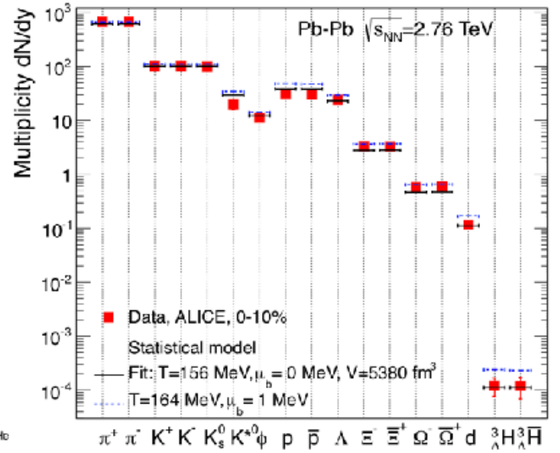
*"Fireballs are made of fireballs, which made of fireballs, which are…"*



- Modification: Regions of hadronization are hydrodynamical in nature
- All regions with the same conserved charges, i.e. $p^{\mu}, Q, S, B$…, will behave equally (In average)
- Hadrons decouple after the crossover to the hadron phase happens



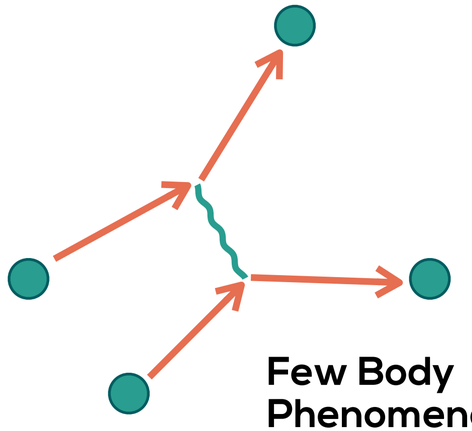




- Data well described, outside of higher mass resonances

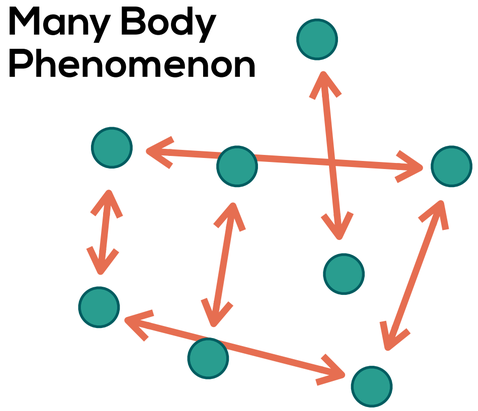
# EMERGENCE

**We are looking for the total, which is “more” than the sum of its parts**



**Few Body Phenomenon**

e.g. 2-2 scattering , bremmstrahlung, etc



**Many Body Phenomenon**

e.g. thermodynamics, plasma dynamics, etc

# Order-of-magnitude physics of the QGP

## Critical temperature at vanishing net baryon number

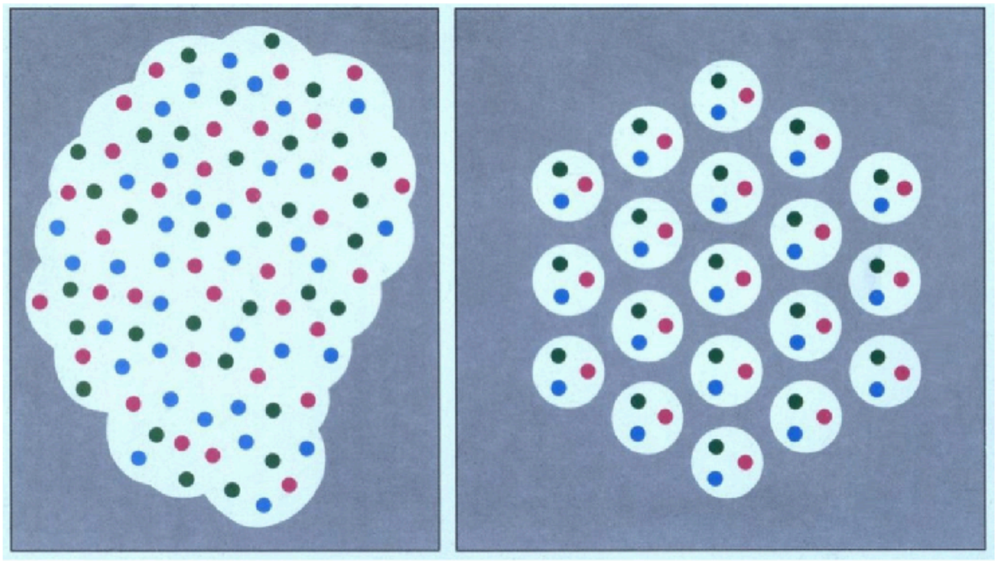
Baryon density of nuclear matter ($R = r_0 A^{1/3}$, $r_0 \approx 1.15$ fm)

$$\rho_0 = \frac{A}{4\pi/3R^3} = \frac{1}{4\pi/3r_0^3} \approx 0.16\,\mathrm{fm}^{-3}$$

Nucleons start to overlap at a critical density $\rho_C$ if nuclear matter is compressed ($r_N \approx 0.8$ fm)

$$\rho_c = \frac{1}{4\pi/3r_n^3} \approx 0.47/\mathrm{fm}^3 = 3\rho_0$$

Figure: CERN



# Order-of-magnitude physics of the QGP

## Critical temperature at vanishing net baryon number

- Consider an ideal gas of u, d quarks and antiquarks, and gluons
- Calculate temperature at which energy density equals that within a proton
- Energy density in a proton

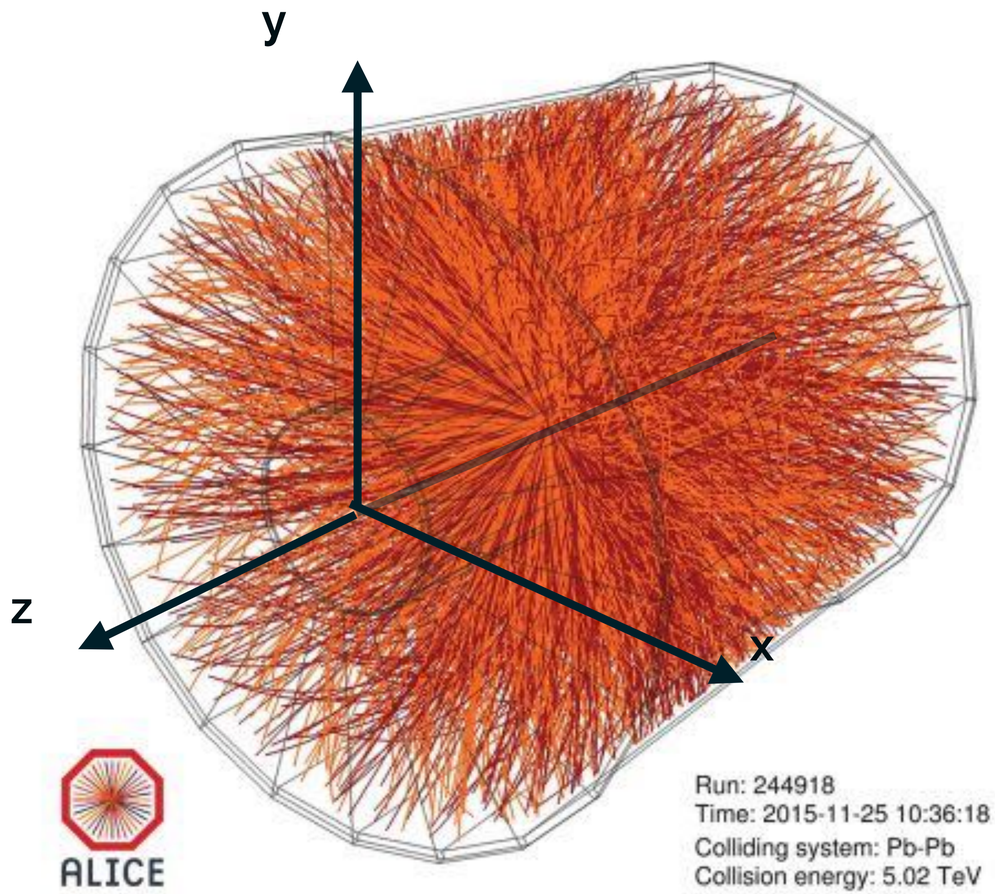
$$\varepsilon_{\text{proton}} = \frac{m}{V} = \frac{0.94\,\text{GeV}}{4/3\pi(0.8\,\text{fm})^3} \approx 0.44\,\text{GeV/fm}^3$$

- Energy density of an ideal gas of massless u and d quarks

$$\varepsilon_{\text{id.gas}} = 37\frac{\pi^2}{30}T^4 = 0.44\,\text{GeV/fm}^3 \rightarrow T \approx 130\,\text{MeV} \qquad (k_B = 1)$$
$$= 1.5 \times 10^{12}\,\text{K}$$

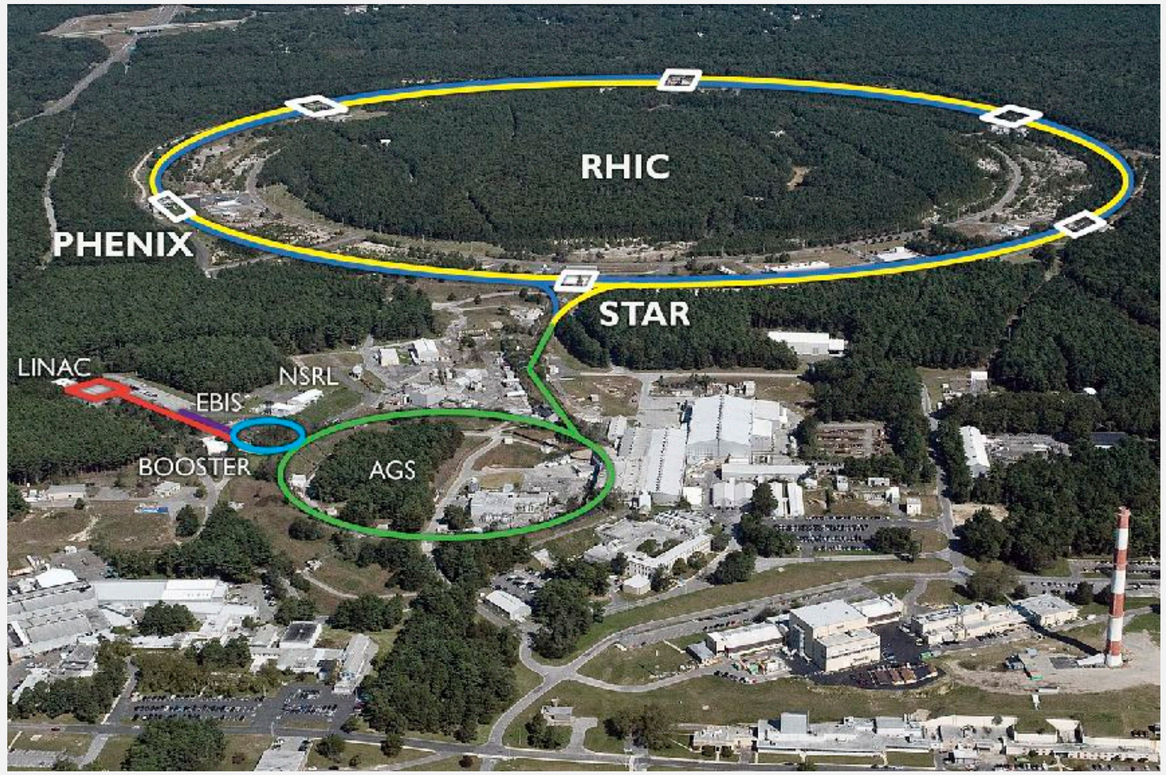
# How can we probe this?

## Heavy Ion Collisions



# Relativistic Heavy Ion Collider

(And the STAR experiment)



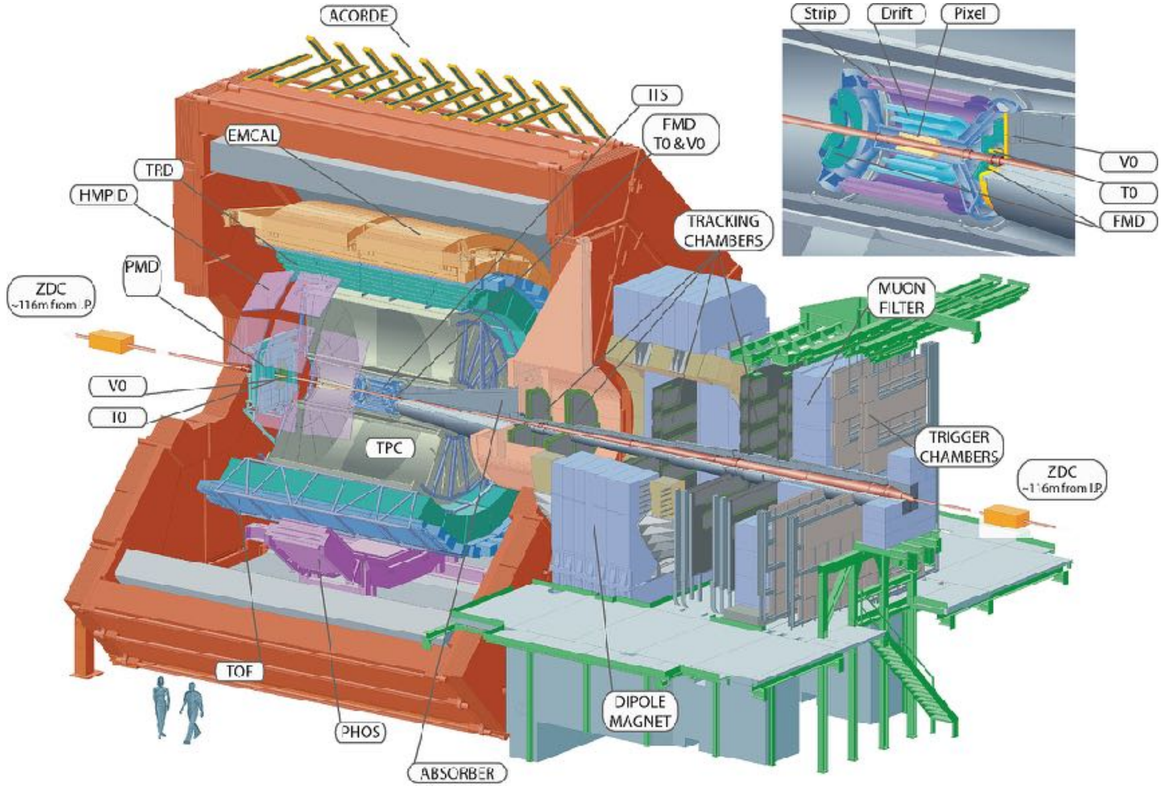



Collision Energy = 200 GeV

# Large Hadron Collider

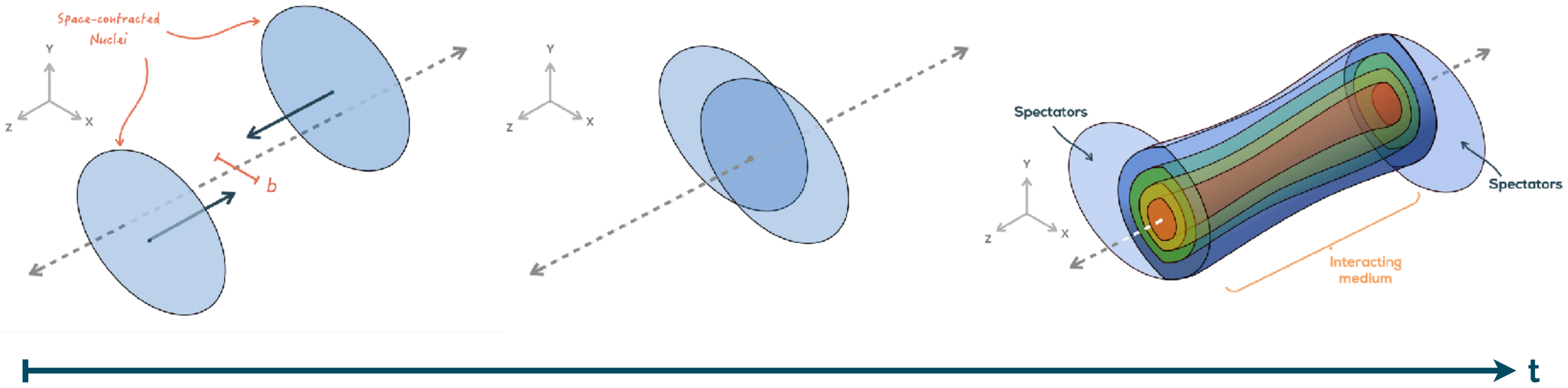
## (And the ALICE experiment)




Collision Energies = 2760, 5020 GeV

# Heavy Ion Collisions: The basic idea




# FAQ

Q: Why heavy-ions?

A: Many initial nucleons, higher energy and number density

Q: Why an interacting medium?

A:We have observed collective dynamics in the resulting spectra

# Heavy Ion Collisions: A cheat sheet.

## The Systems

p+p → Baseline ("Control Group")

p+A → Cold Nuclear Matter lab

A+A → Hot Nuclear Matter lab

A=C, O, Au, Pb, …, U

## What we can observe

### Invariant Yields

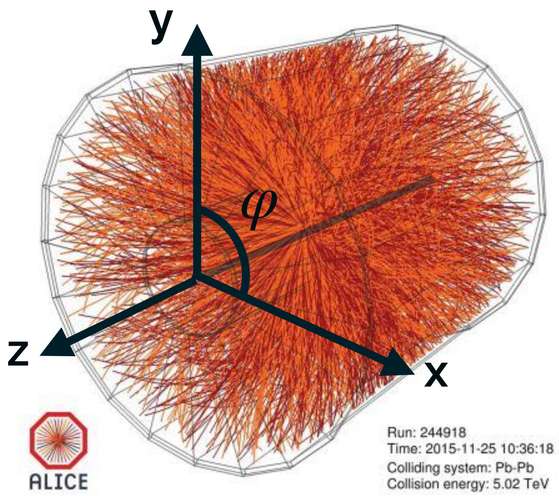
$$\frac{\mathrm{d}N}{\mathrm{d}y\mathrm{d}^2p_\perp}$$

- Yields: $\mathrm{d}N/\mathrm{d}y$
- $p_\perp$ spectrum: $\mathrm{d}N/\mathrm{d}p_\perp$
- Angular spectrum: $\mathrm{d}N/\mathrm{d}\phi$

### Particle Correlations

$$\frac{\mathrm{d}N}{\mathrm{d}y_1\mathrm{d}^2p_\perp\mathrm{d}y_2\mathrm{d}^2q_\perp}$$

- Interferometry functions
- Cumulants



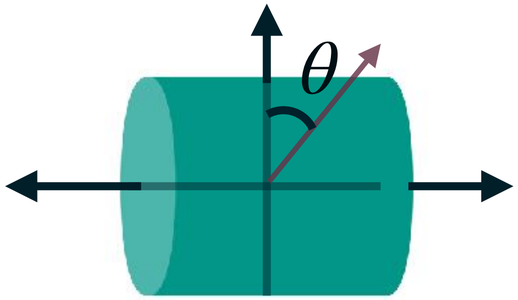

## The variables we use

Momentum rapidity $y = \frac{1}{2}\log\left[(E+p_z)/(E-p_z)\right]$

$\sim \theta$ if massless…

Transverse momentum $\mathbf{p}_\perp = (p_x, p_y)$

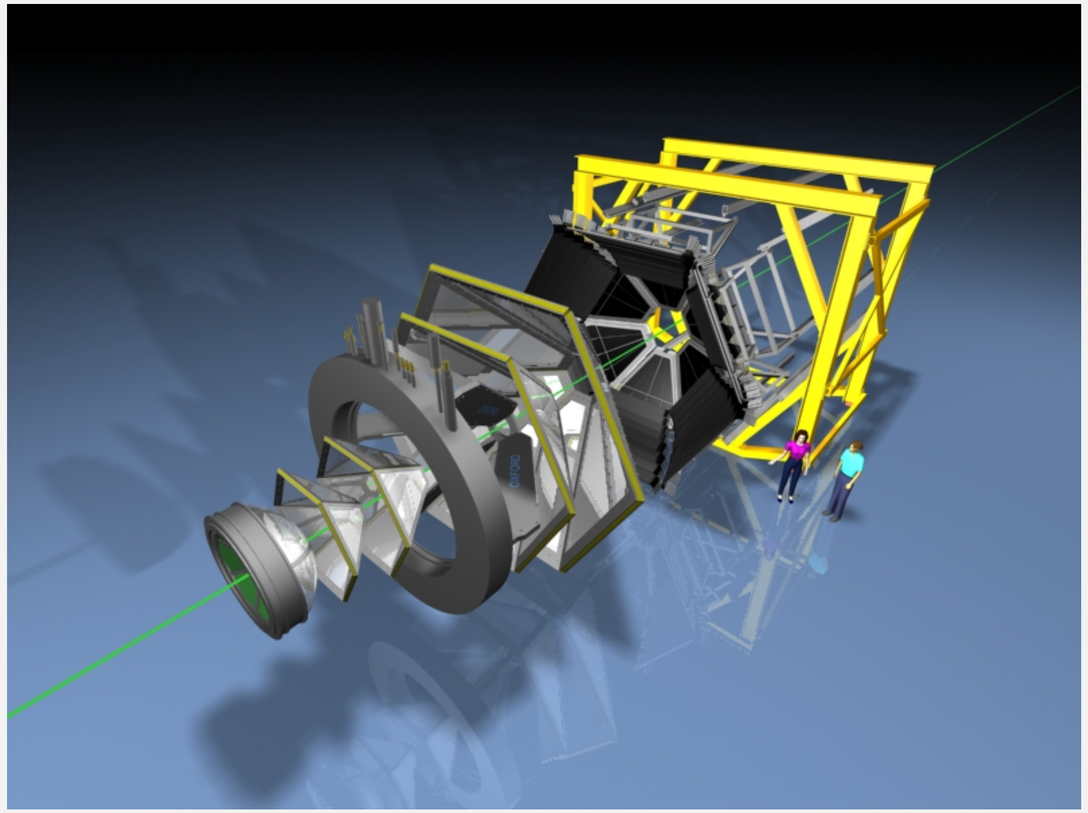
Center of mass energy. $s = (p_1 + p_2)^2$



# SIS18 at GSI

## (And the HADES experiment)





Collision Energy = 1-5 GeV

# I. Cosmology

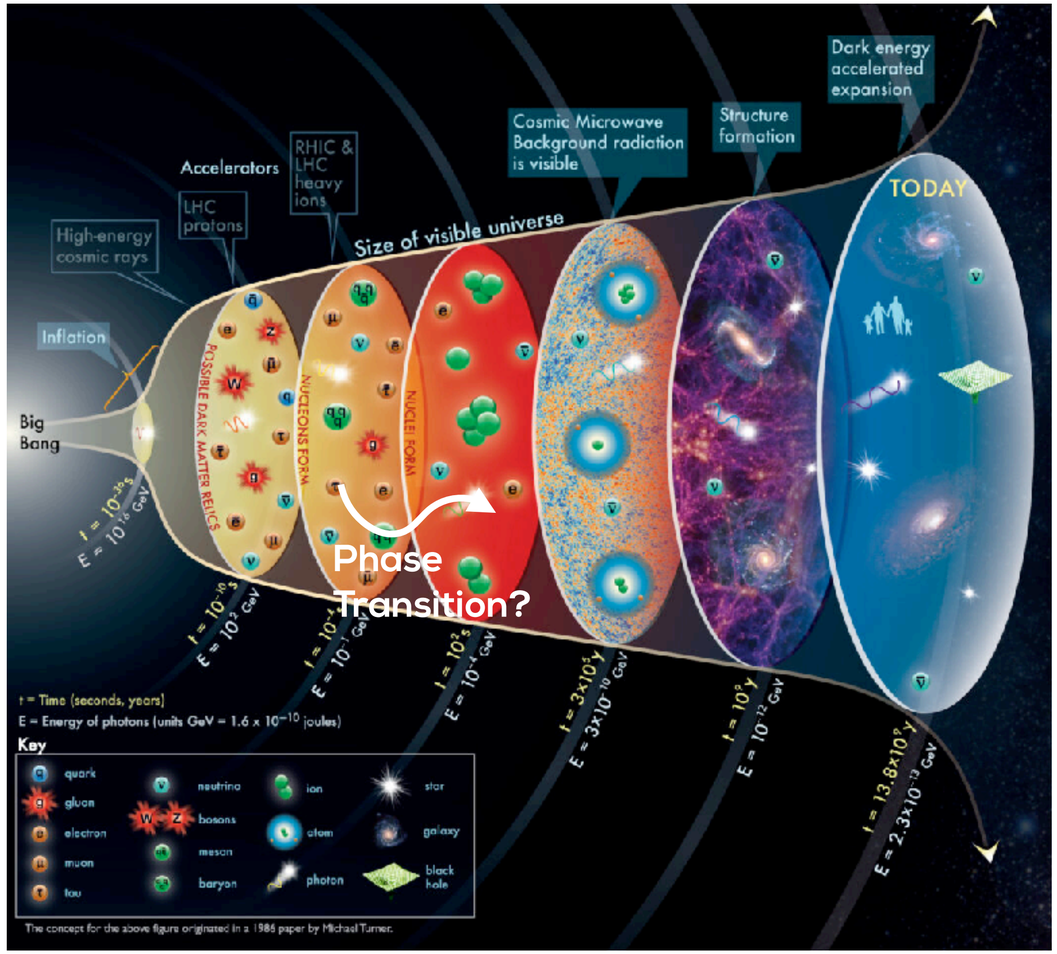



Image from CPEPphysics.org

Energy density in the Early Universe ($10^{-3}$ s) should be high enough to *"melt"* baryon and meson states

New deconfined state would be a deconfined, collective plasma of quarks and gluons:

**The Quark-Gluon Plasma**

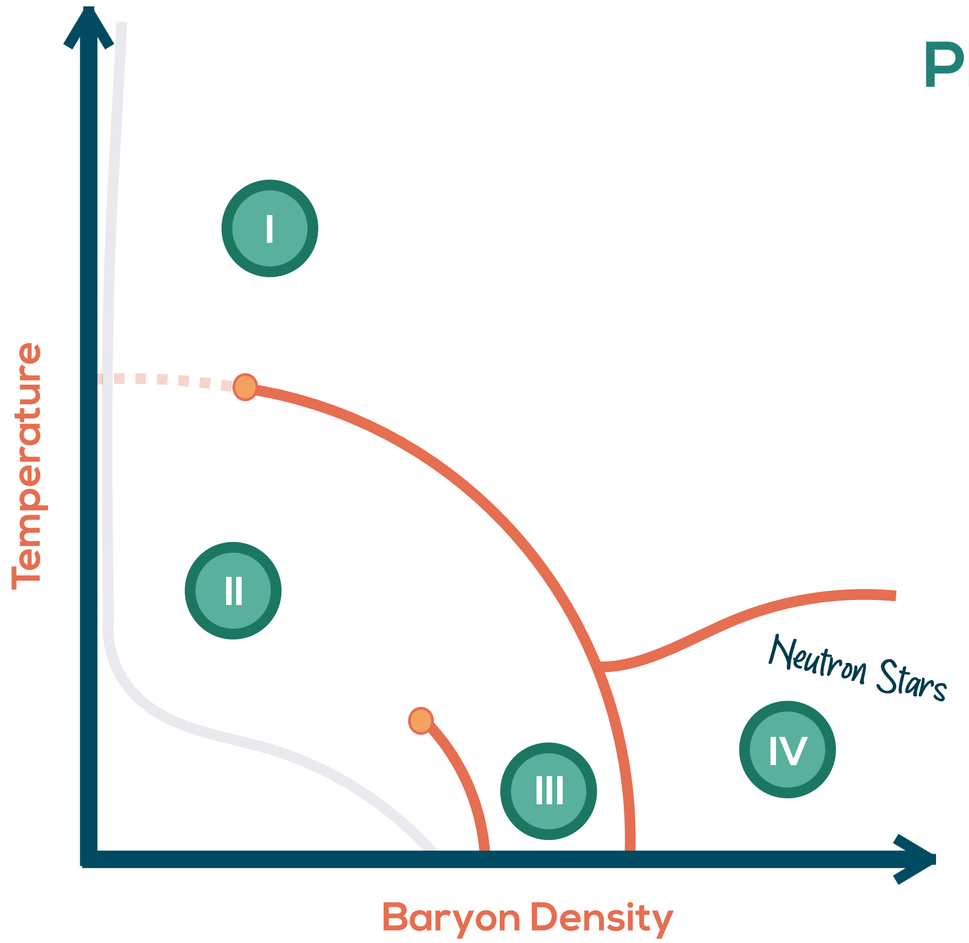
Idea: with expansion cooling, a phase transition must to hadrons must have happened.

- If 1st or 2nd order PT order phase would leave imprint on the universe.

No effect found in data… Crossover?

# II. Phase Diagram

Interest in understanding the structure of nuclear matter for different conditions.

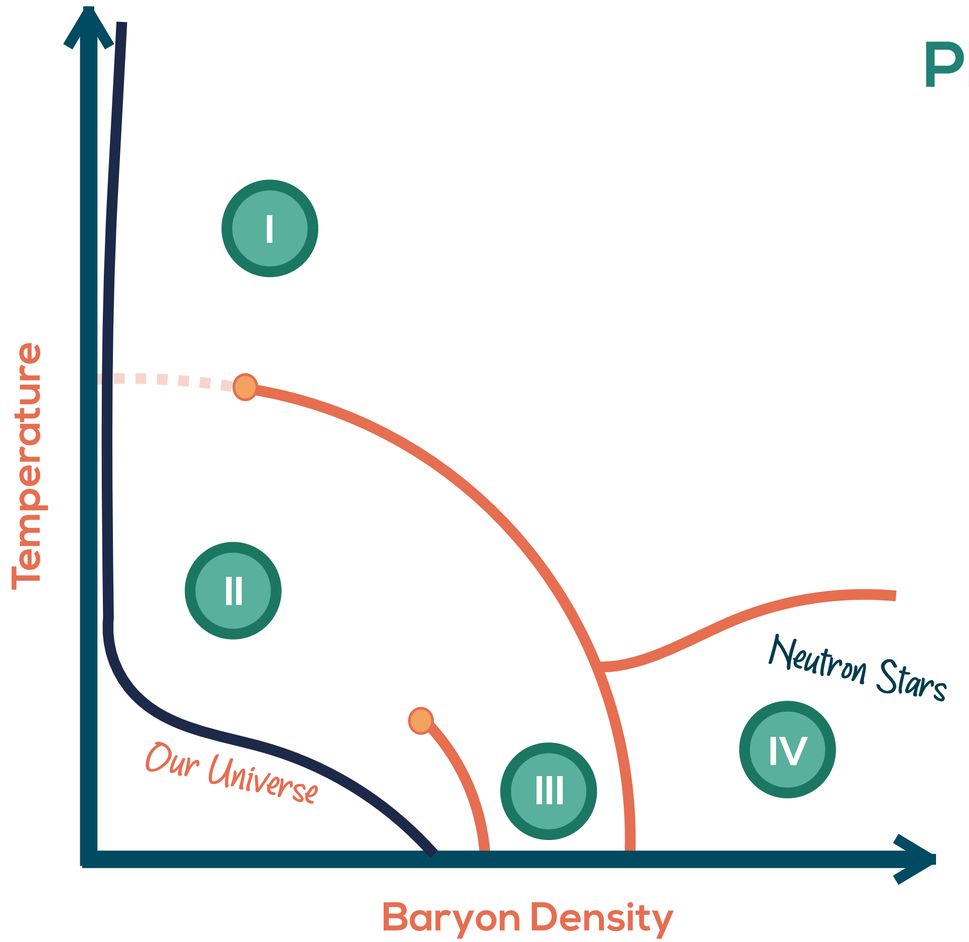



## Phase Diagram of Nuclear Matter

- I Quark-Gluon Plasma
- II Hadronic Gas
- III Normal Matter
- IV "Exotic States"
  - Nuclear Pasta
  - Color Superfluid
  - Color Superconductor

# II. Phase Diagram

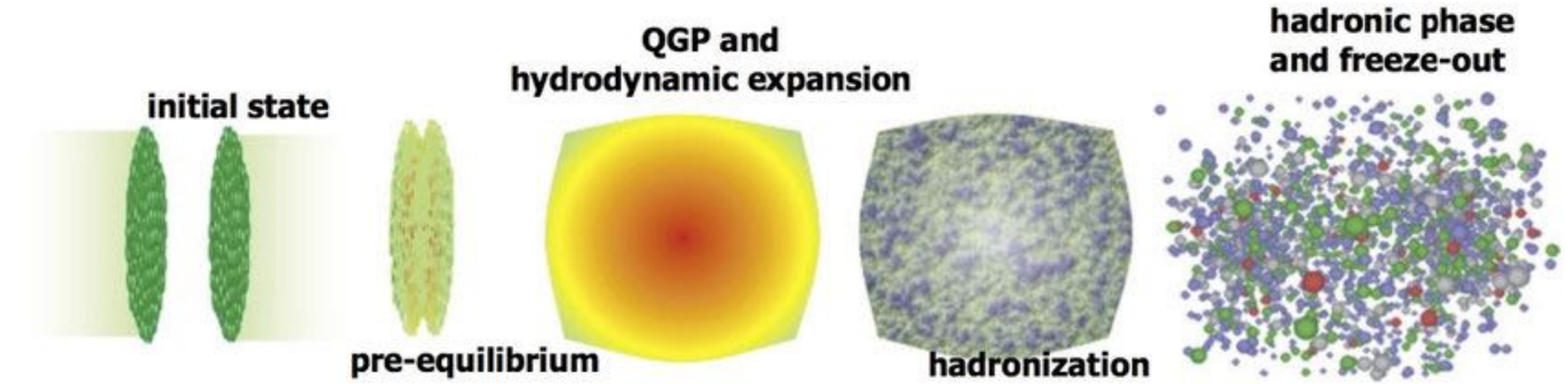
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## Phase Diagram of Nuclear Matter

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  - Color Superconductor

# *The Standard Model of Heavy Ion Collisions*




$\tau = 0$ $\tau \sim 1\,\mathrm{fm}$ $\tau \sim 6 - 10\,\mathrm{fm}$ $\tau \sim 20 - 50\,\mathrm{fm}$

# ALL MODELS ARE WRONG

BUT SOME ARE USEFUL...

# The initial State: When, what and how?

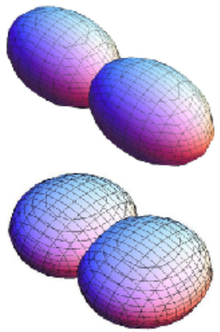
**Initial State Modeling**: *"Finding the correct number and density of conserved charges, and their **spatial distribution**"*

**Initial state fluctuations**

**Geometric**

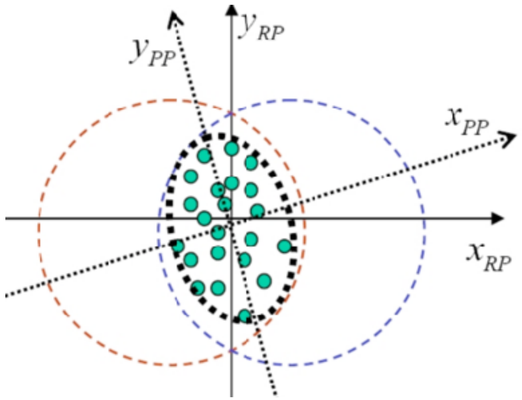
**Deformation**

Nucleus is not spherical (prolate or oblate)



**Collisional**

Collisions generate geometrical anisotropies

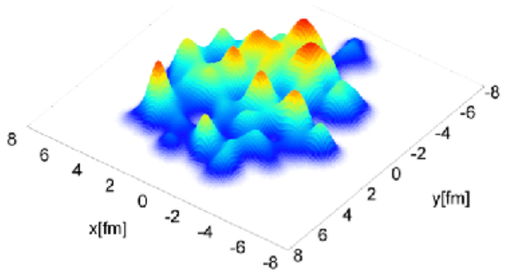



**Dynamical**

**Nucleonic**

Energy and charge density fluctuations at the level of nucleons

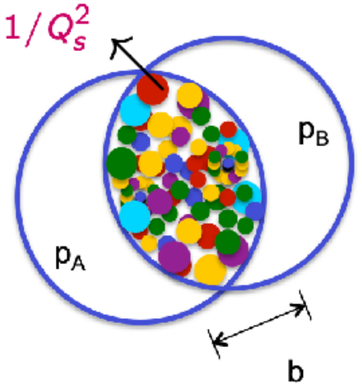
$d \sim R_p$




**Sub-nucleonic**

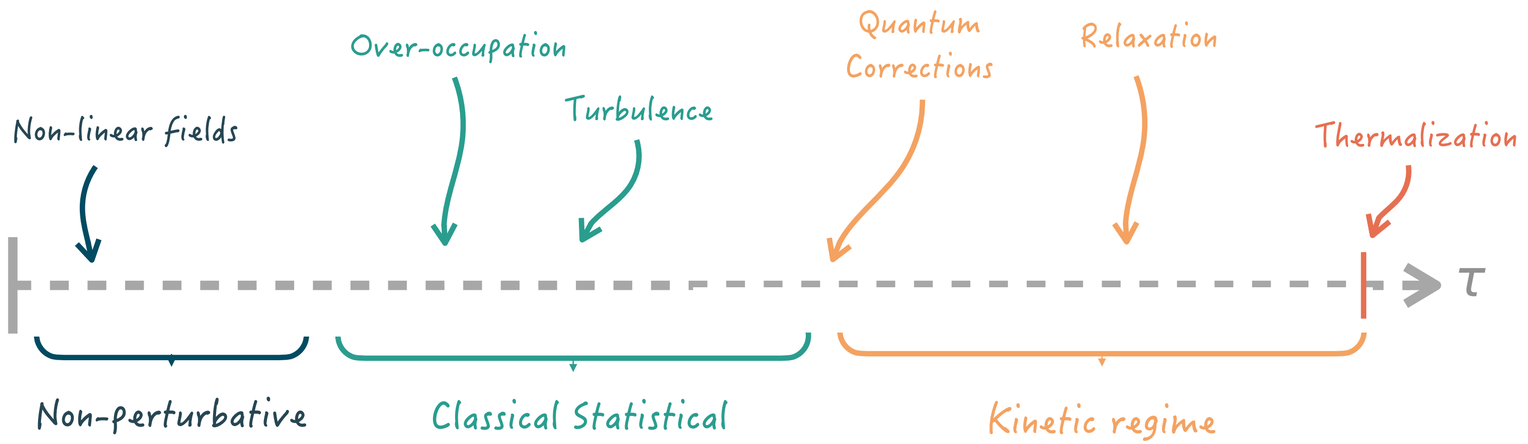
Energy and charge density fluctuations at the level of partons

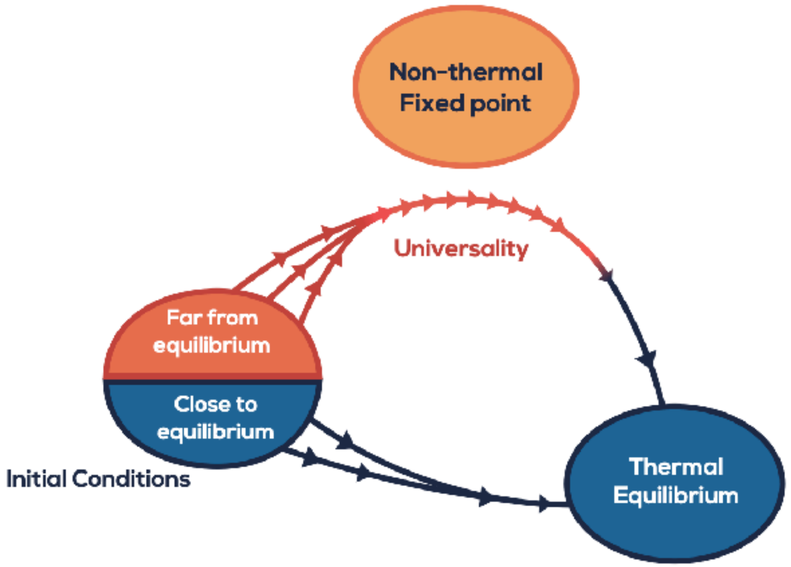
$d \sim 1/Q_S$



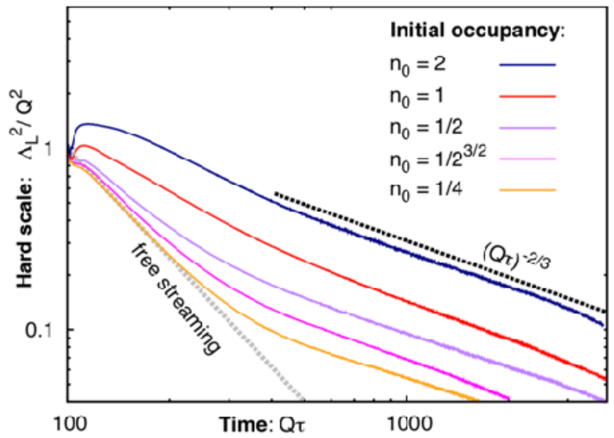
# Pre-equilibrium Dynamics: Turbulent Thermalization

How does one go from an extremely violent event turn into a nicely thermalized fluid?

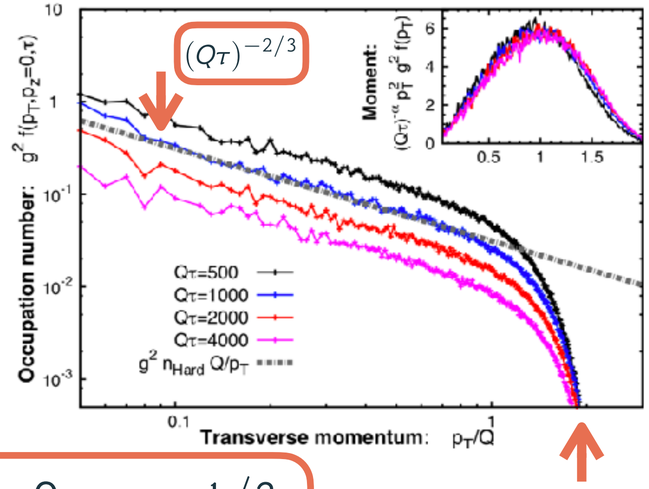



**Hard Scale:** $\Lambda_L^2 \sim \langle p_z \rangle^2$




**Transverse** $p_\perp$




$$\alpha = -2/3 \quad \beta = 0 \quad \gamma = 1/3$$

## Non-thermal fixed point

def. Parametrically long self-similar regime quantum fields under go in their way to Thermal Equilibrium

## Self - Similarity

def. distribution function depends on a Universal, time-independent function

$$f(t,p) = t^{\alpha} f_S(p_{\perp} t^{\beta}, p_z t^{\gamma})$$






## Transport and Turbulence

def. Local flow of conserved charges to accommodate better the total corresponding charge. The flow is turbulent when is self-similar

# QGP stage: the hottest fluid in existence

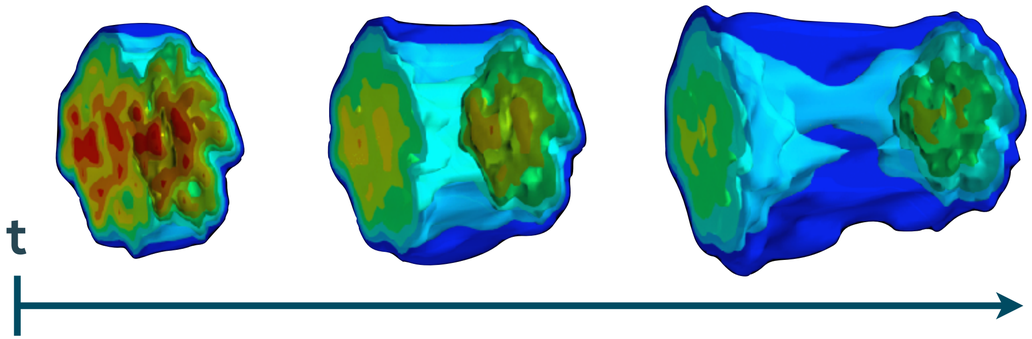
Another FAQ:

Q: Why is a system with some O(100) particles hydrodynamical?

A: Uff, tough question, but! The nucleon assessment is not the full story → Parton density!

Q: What characterizes a hydrodynamical system

A: Hydro is conservation of charges.




$$\partial_\mu T^{\mu\nu} = 0 \qquad \text{and} \qquad \partial_\mu N_i^\mu = 0 \quad \text{with} \quad i = Q, B, S$$

Energy density

Rest frame pressure

Heat flux current $u_\mu q^\mu = 0$

Shear stress tensor $\pi^\mu{}_\mu = 0$ $u_\mu \pi^{\mu\nu} = 0$

$$T^{\mu\nu} = \epsilon u^\mu u^\nu - (p + \Pi)\,\Delta^{\mu\nu} + 2\,q^{(\mu} q^{\nu)} + \pi^{\mu\nu}$$

Fluid 4-velocity $u^\mu u_\mu = 1$

Bulk viscous pressure $3(p + \Pi) = -\Delta^{\mu\nu} T_{\mu\nu}$

Projector $u_\mu \Delta^{\mu\nu} = 0$

Too many degrees of freedom!
Hydro needs extra knowledge
(assumptions to close the system)

Equation of state $P = P(E, S, N_i)$

Bulk Pressure $\dot{\Pi} = F_{(\zeta,T)}[\Pi, u]$

Shear Stress $\dot{\pi} = F_{(\eta,T)}[\pi, u]$

# QGP stage: the hottest fluid in existence

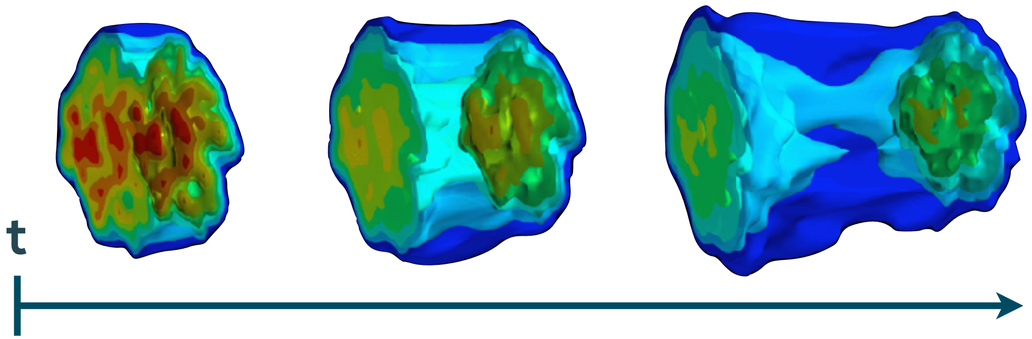
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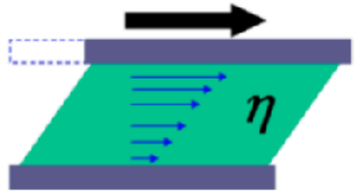
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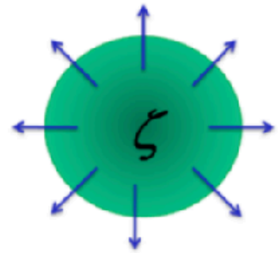
Projector $u_\mu \Delta^{\mu\nu} = 0$

## Transport coefficients



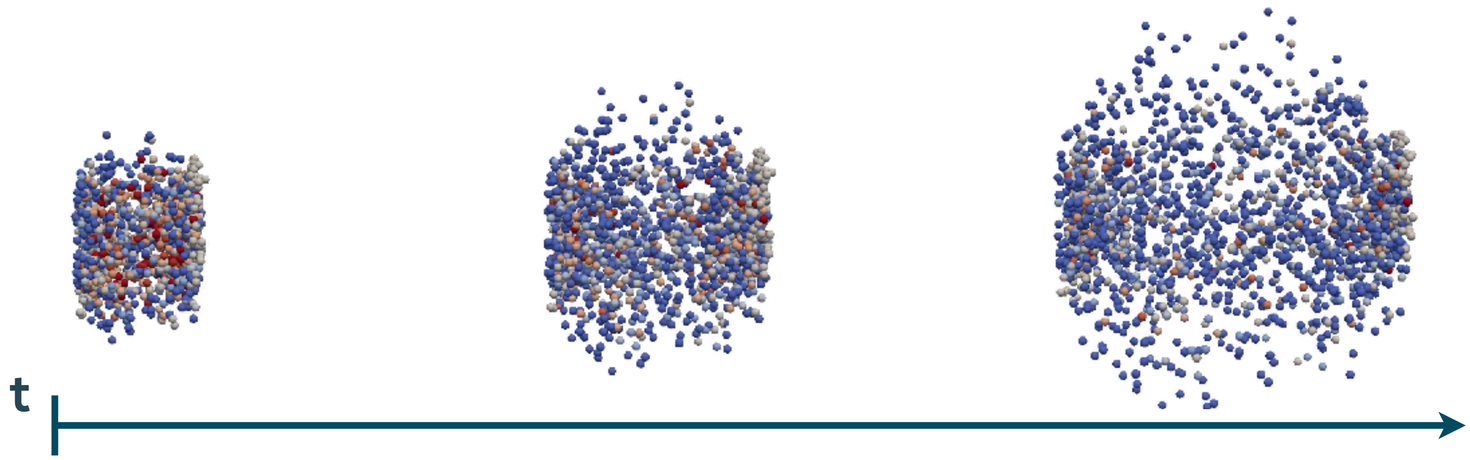

**Shear Viscosity**
Resistence to flow and ability of momentum transfer

**Bulk Viscosity**
Resistence to isotropic expansion and dynamics of compressibility of the fluid



# The hadronic stage

- After hadronization, one can continue modelling the evolution as a hydrodynamical fluid, or use hadrons as microscopic DoFs.
- Evidence suggests out-equilibrium effects (hadronic rescattering) are important to match data.
- Hadronic non-hydro effects are included using Afterburner Codes
- Particles propagate and collide on the basis of physical cross sections
- Effective solution to Boltzmann equation: $p_\mu \partial^\mu f + m \partial_{p_\mu}(F^\mu f)^\mu = C[f]$
- Hadronic transport codes in our collaboration: (SMASH, UrQMD …)



# What can we learn from hot deconfined QCD?

**QCD Thermodynamics**

Equation of state $\epsilon, p, s$

Speed of sound $c_s^2 = \partial p / \partial \epsilon$

Shear Viscosity $\eta$

Bulk Viscosity $\zeta$

Momentum, energy and charge diffusion coefficients $\hat{q}, \hat{e}, \hat{e}_2$

Color screening mass $m_D^2$

Affects →

← We can learn/extract

**Actual measureable quantities**

Spectra, collective flow, *particle* fluctuations

Particle correlations

Anisotropic flow

Hadronic average momentum

Parton energy loss/jet fragmentation

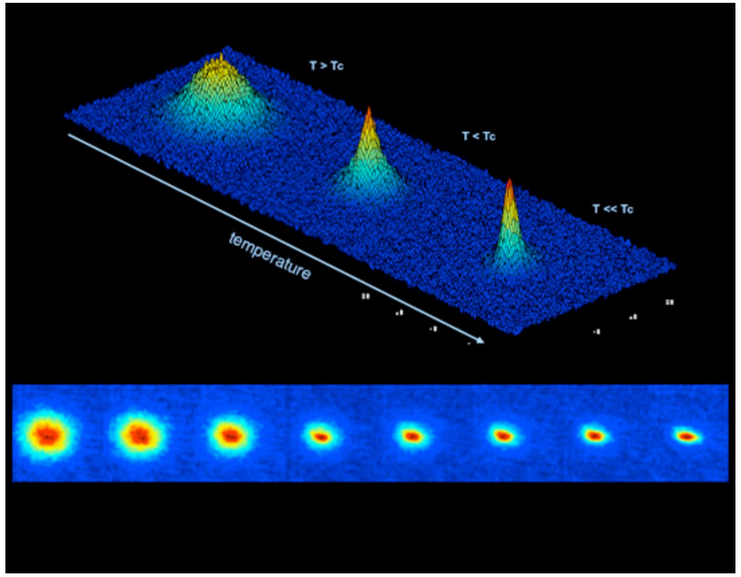
Abundance of $q\bar{q}$ (quarkonic) mesonic states

# Caveat: Complex Quantum Matter

Heavy ion collisions are creating isolated quantum systems.

Testing the construction of complex quantum matter in the high energy frontier.

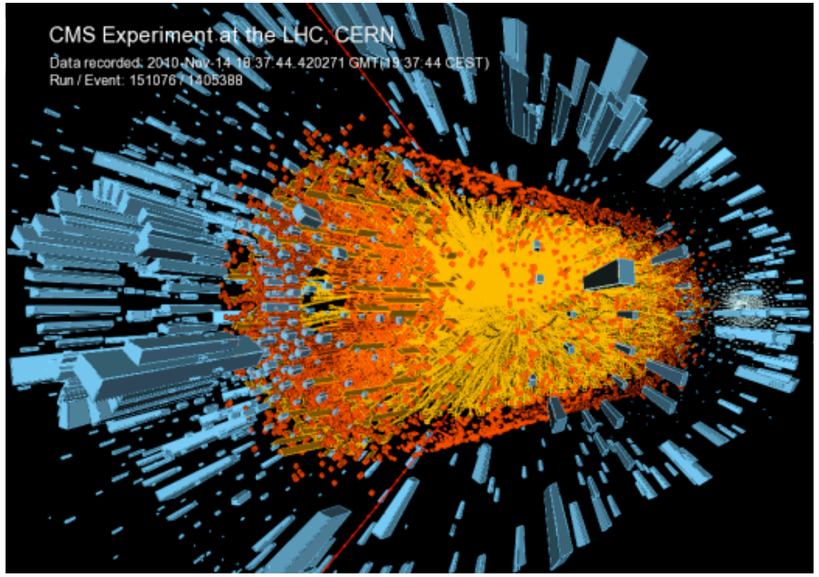
Many interesting questions: Entanglement, Non-equilibrium dynamics and thermalization, dynamical phase transitions, and so much more.




Ultracold Quantum Gases, production of BEC

Understanding quantum dynamics

Throughout scales and many, many orders of magnitude



Now some of the promised

# OBSERVABLES

of collective, emergent dynamics

# Πάντα ῥεῖ - Everything Flows

- Angular anisotropies found in particle spectra produced in Nucleus-Nucleus collisions

$$\frac{\mathrm{d}N}{\mathrm{d}^2p_\perp \mathrm{d}y} = \frac{1}{2\pi\, p_\perp}\frac{\mathrm{d}N}{\mathrm{d}p_\perp \mathrm{d}y}\left\{1 + 2\sum_i v_i(p_\perp)\cos\left[n(\phi - \Phi_{ev})\right]\right\}$$

(Just a run-of-the-mill Fourier expansion…)

- Flow dynamics signals collectivity, medium creation

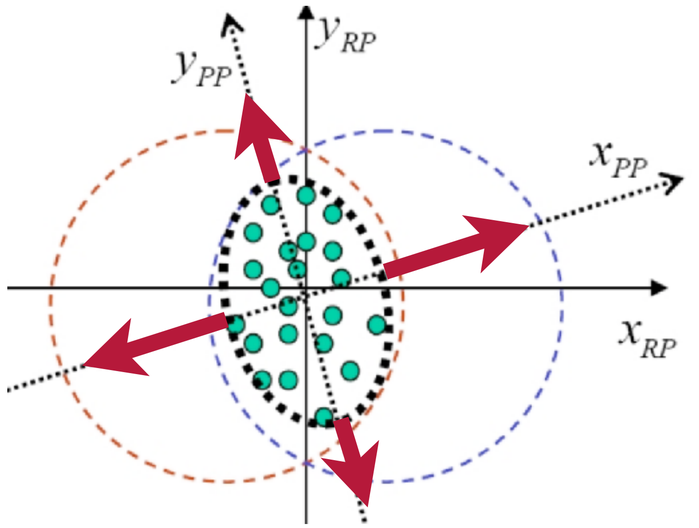
- Why is does this mean collective dynamics?

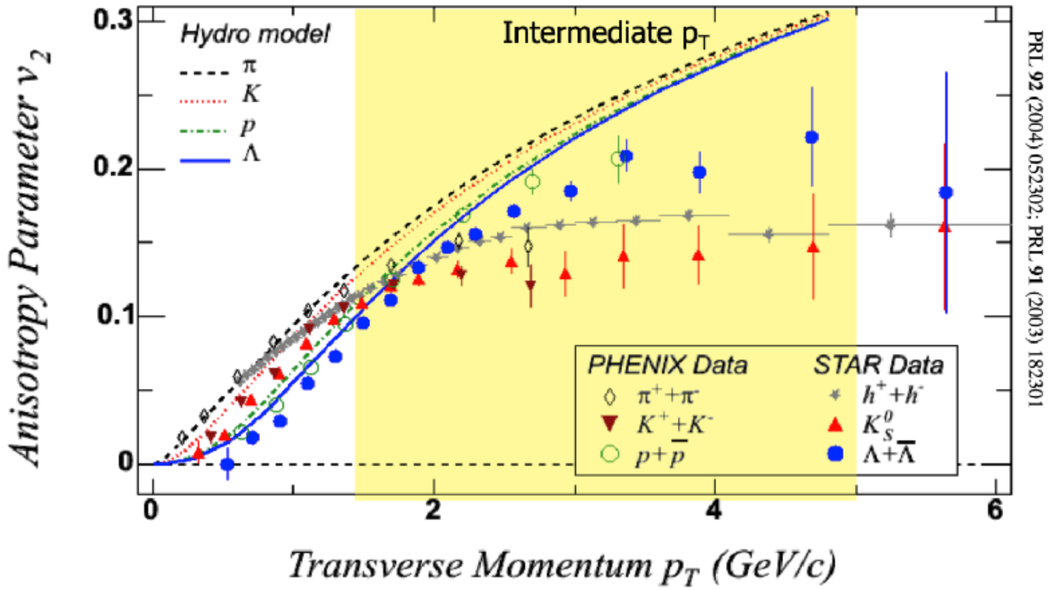
Interactions

$$\varepsilon = \frac{\langle y^2 - x^2\rangle}{\langle y^2 - x^2\rangle} \qquad v_2 = \frac{\langle p_y^2 - p_x^2\rangle}{\langle p_y^2 - p_x^2\rangle}$$

Spatial anisotropy (eccentricity)

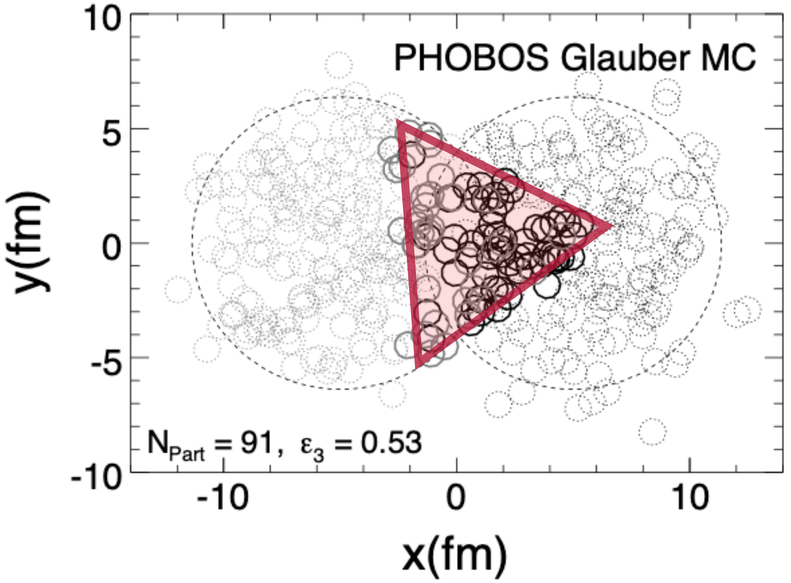
Momentum anisotropy

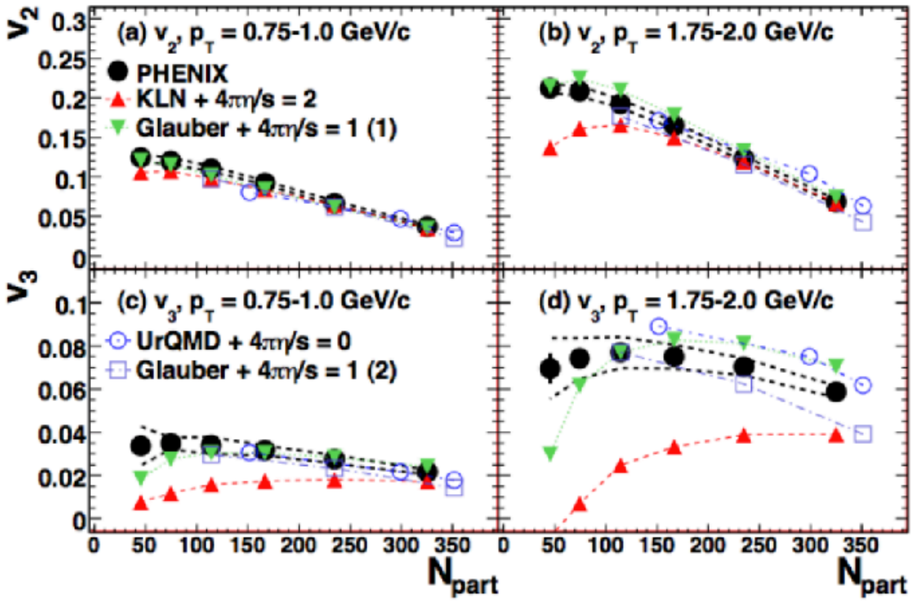


# The initial State: Geometry and Correlations?




Higher flow coefficients have been found to be non-vanishing at RHIC and LHC energies.

What can we learn from these observables?

$$\varepsilon = \frac{\langle y^2 - x^2 \rangle}{\langle y^2 - x^2 \rangle} \quad \xrightarrow{\text{Interactions!!!}} \quad v_2 = \frac{\langle p_y^2 - p_x^2 \rangle}{\langle p_y^2 - p_x^2 \rangle}$$

Spatial anisotropy (eccentricity)

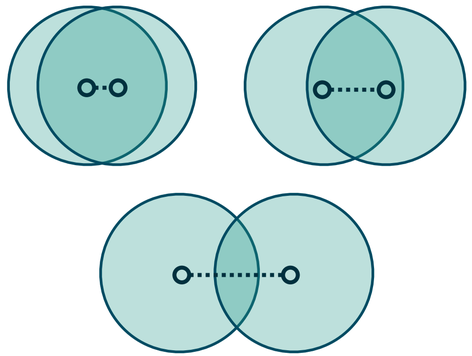
Momentum anisotropy

# A little intermezzo: Two useful tools.

## Centrality

Organizational variable to organize events by number of particles created $\mathrm{d}N_{ch}/\mathrm{d}y$

Physical interpretation: overlap area of collision. Basically volume dependence



How to read it?
"A 10-20% observable is the collection of events between the 10 and 20% of the largest particle output"

## Nuclear Modification Factor

Way to isolate and compare nuclear effects using the p+p *baseline.*

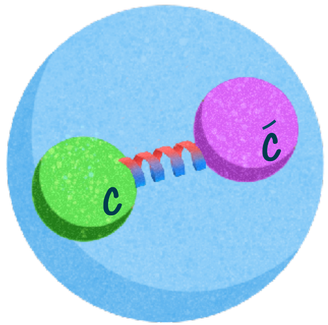
$$R_i = \frac{\mathrm{d}N_i/\mathrm{d}^2p_\perp\mathrm{d}y}{N_{coll}\,\mathrm{d}N_{pp}/\mathrm{d}^2p_\perp\mathrm{d}y} \quad \text{with} \quad i = \{pA, AA\}$$

One needs to rescale with respect $N_{coll}$, the average number of nucleonic collisions expected from the area overlap.

If $R_i = 1$ , then the nuclear collision is a collection of incoherent (uncorrelated) collisions

# Quarkonia Suppression

- In-medium screening reduces strength of quark-antiquark bonding
- Suppression is used as a “thermometer"




J/Psi

$$R_i = \frac{\mathrm{d}N_i/\mathrm{d}^2p_\perp \mathrm{d}y}{N_{coll}\,\mathrm{d}N_{pp}/\mathrm{d}^2p_\perp \mathrm{d}y}$$

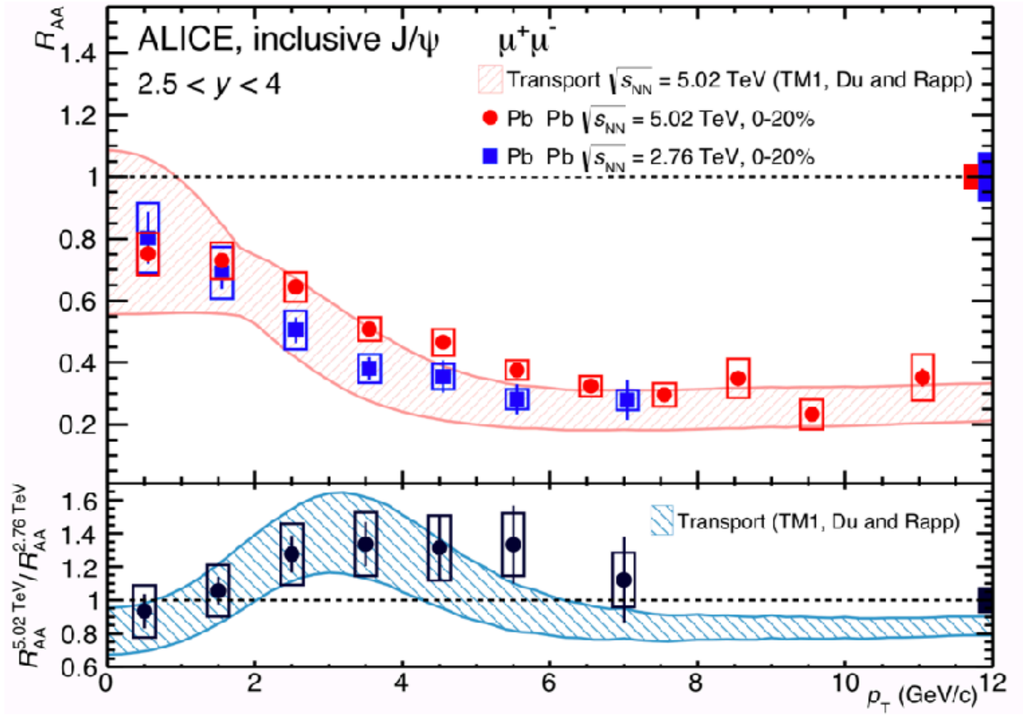



Figure from ALICE collaboration, Phys.Lett. B766 (2017) 212-224

# Quarkonia Suppression

- In-medium screening reduces strength of quark-antiquark bonding
- Here in this example, we observe sequential suppression on $\Upsilon$ states.
- Resonances are larger and more weakly bound
- Checking relative suppression of higher mass states can give hints at the temperature of the QGP

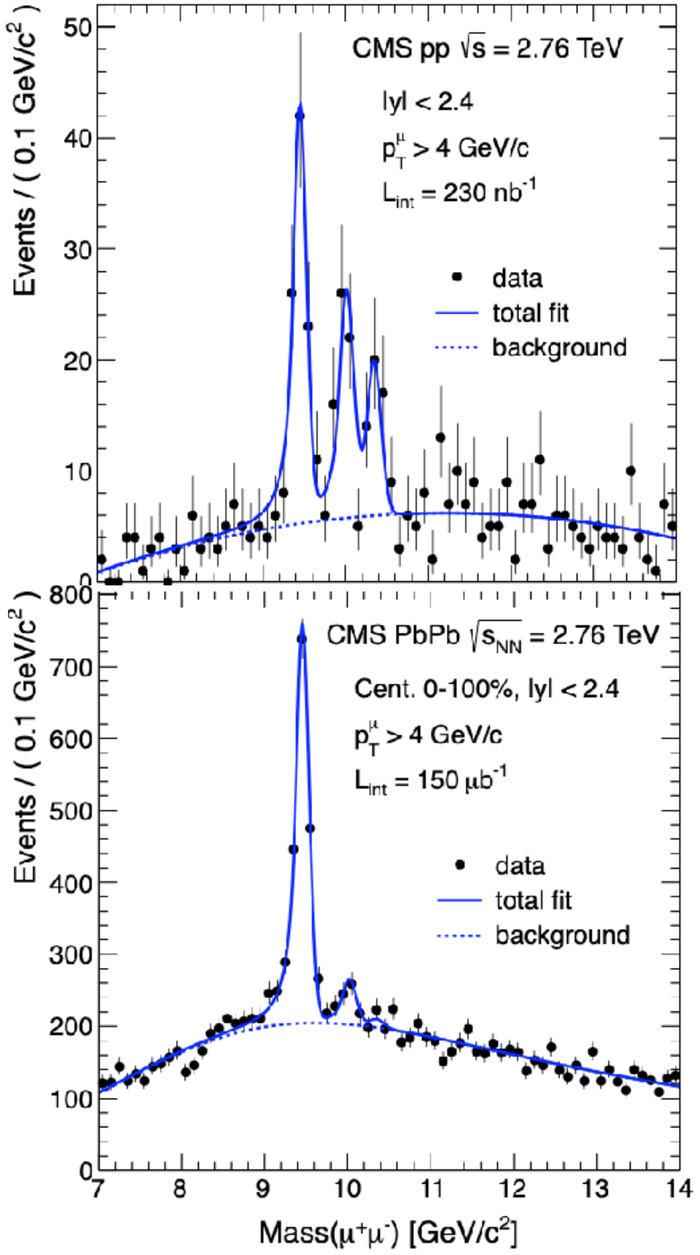
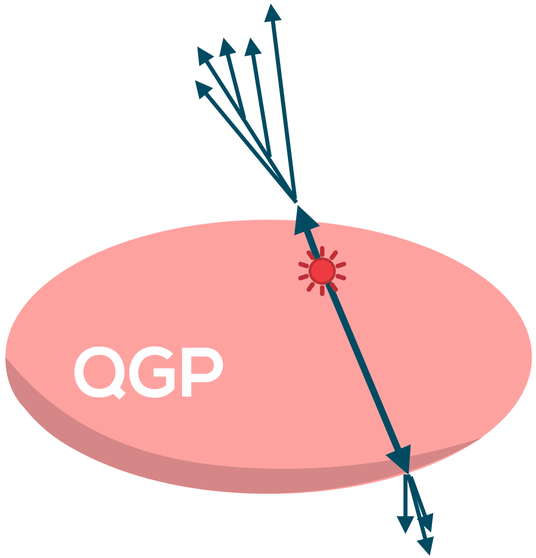


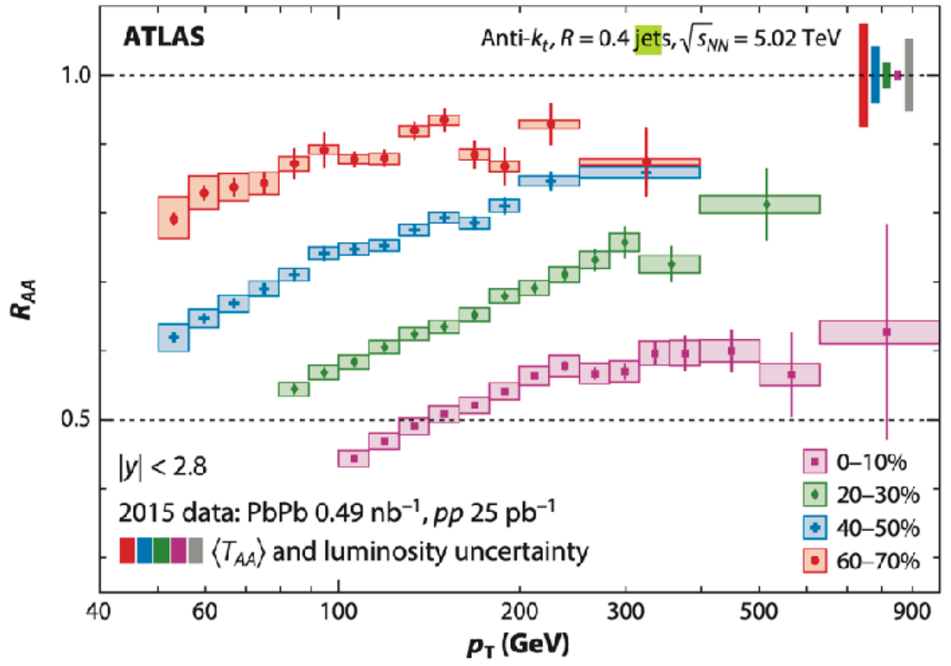

Figure from CMS collaboration, Phys.Rev.Lett. 109 (2012) 222301

# Jet Quenching

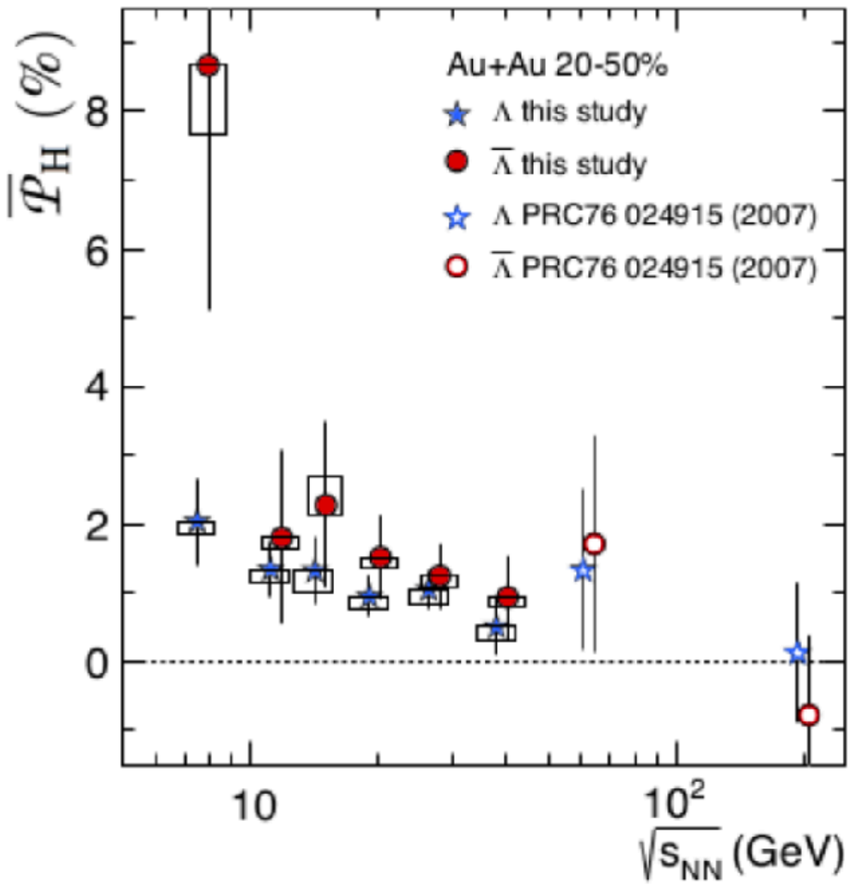


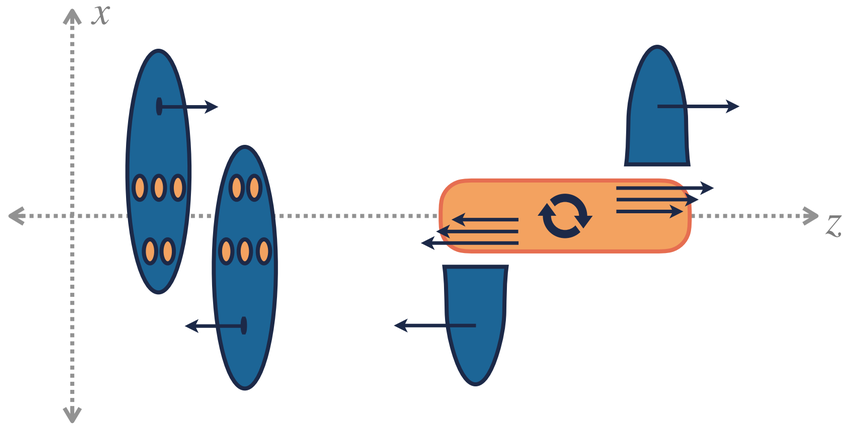

- Jets are collimated influx of particles produced by highly energetic energetic initial collisions.
- With the production of a medium, jet looses energy to the plasma.
- In vacuum jets are symmetric. Medium breaks symmetry.
- Nuclear Modification Factor is centrality dependent, which is pretty telling

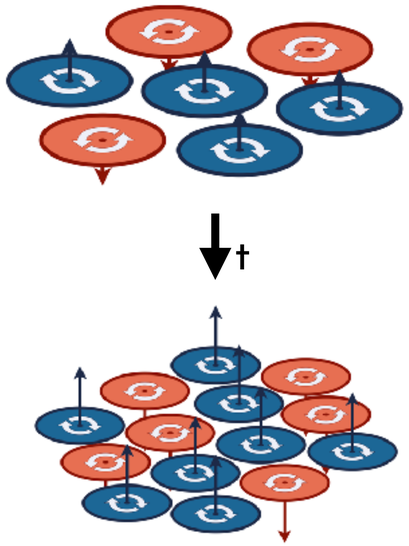
$$R_i = \frac{\mathrm{d}N_i/\mathrm{d}^2p_\perp \mathrm{d}y}{N_{coll}\,\mathrm{d}N_{pp}/\mathrm{d}^2p_\perp \mathrm{d}y}$$



# Polarization of $\Lambda$ Particles

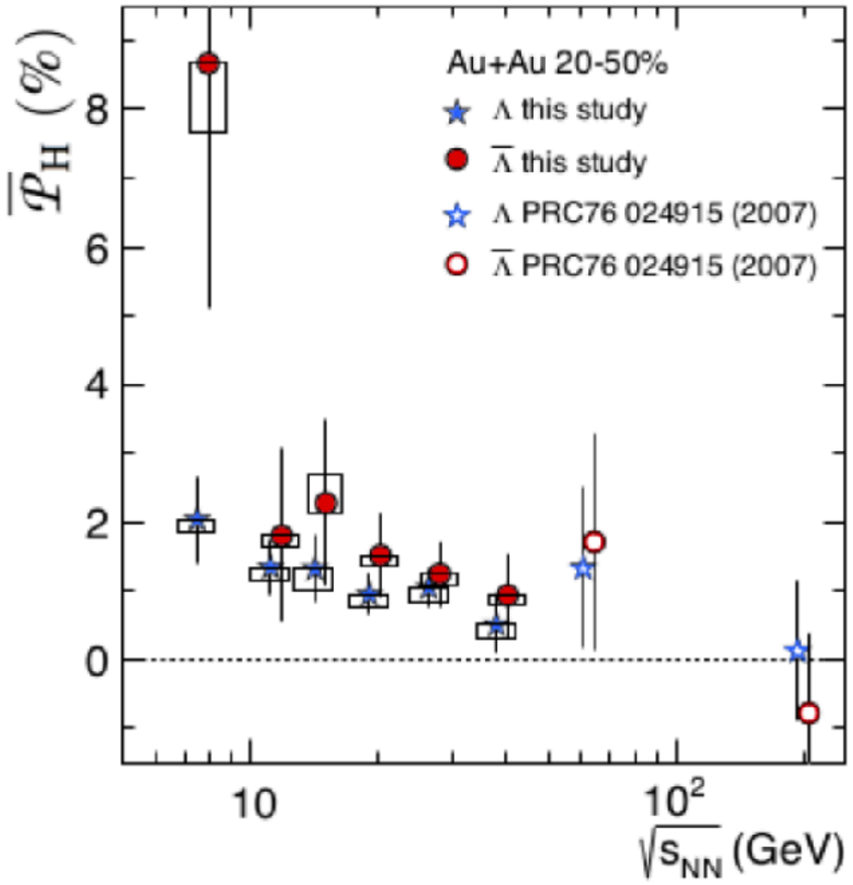


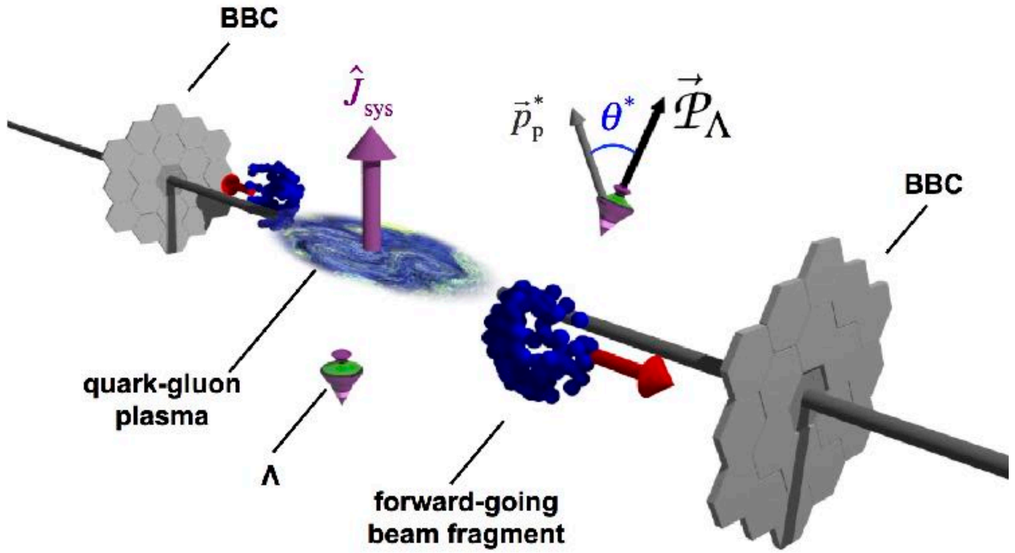




- $\Lambda$ Initial huge angular momentum in the collision needs to be redistributed in the form of vorticity
- $\Lambda$ Vorticity pockets induce local rotation. Spin-1/2 particles align to these pockets
- $\Lambda$ Polarization observed stronger at lower energies
- $\Lambda$ Vortically induced polarization tells us that the effect is thermal-mechanical. A good hint that a medium is produced.

# Polarization of Λ Particles







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- Λ Vorticity pockets induce local rotation. Spin-1/2 particles align to these pockets
- Λ Polarization observed stronger at lower energies
- Λ Vortically induced polarization tells us that the effect is thermal-mechanical. A good hint that a medium is produced.

# If we want to learn about deconfined QCD, we need dynamical modeling

**QCD Thermodynamics**

Equation of state $\epsilon, p, s$

Speed of sound $c_s^2 = \partial p / \partial \epsilon$

Shear Viscosity $\eta$

Bulk Viscosity $\zeta$

Momentum, energy and charge diffusion coefficients $\hat{q}, \hat{e}, \hat{e}_2$

Color screening mass $m_D^2$

Affects →

← We can learn/extract

**Actual measureable quantities**

Spectra, collective flow, *particle* fluctuations

Particle correlations

Anisotropic flow

Hadronic average momentum

Parton energy loss/jet fragmentation

Abundance of $q\bar{q}$ (quarkonic) mesonic states

# OTHER (COOL) QUESTIONS

# Quantum Dynamics throughout scales

Other isolated quantum systems seem to behave similar to our QGP!

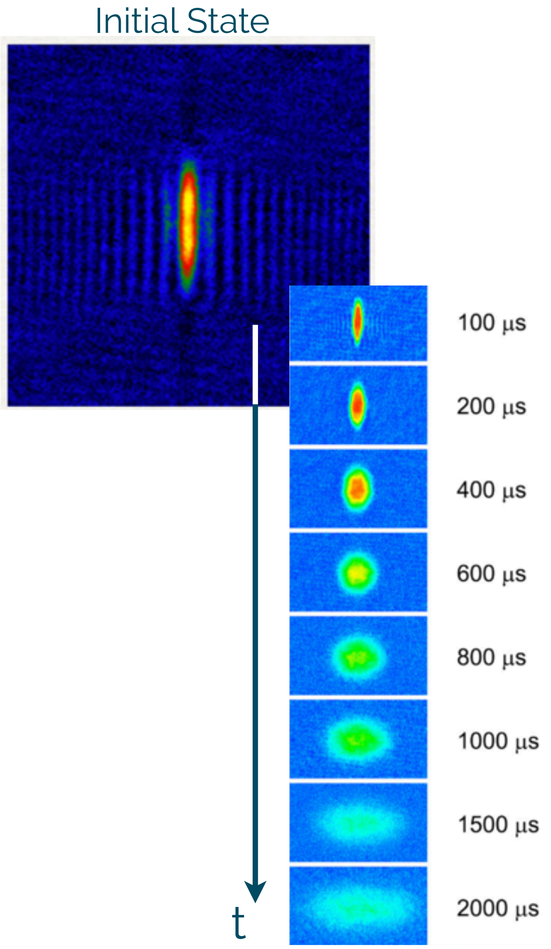
Hottest and coldest matter on earth exhibits dynamics universal to QCD at very large energies and temperatures

First example:

Elliptic flow build-up in cold quantum gas experiments.

Lithium atoms released from an optical trap in a cigar (almond)-shape

Hydro-like gradients create anisotropies through the interactions of the relevant degrees of freedom!




O'Hara *et al*. Science **298** (2002) 2179

# Quantum Dynamics throughout scales

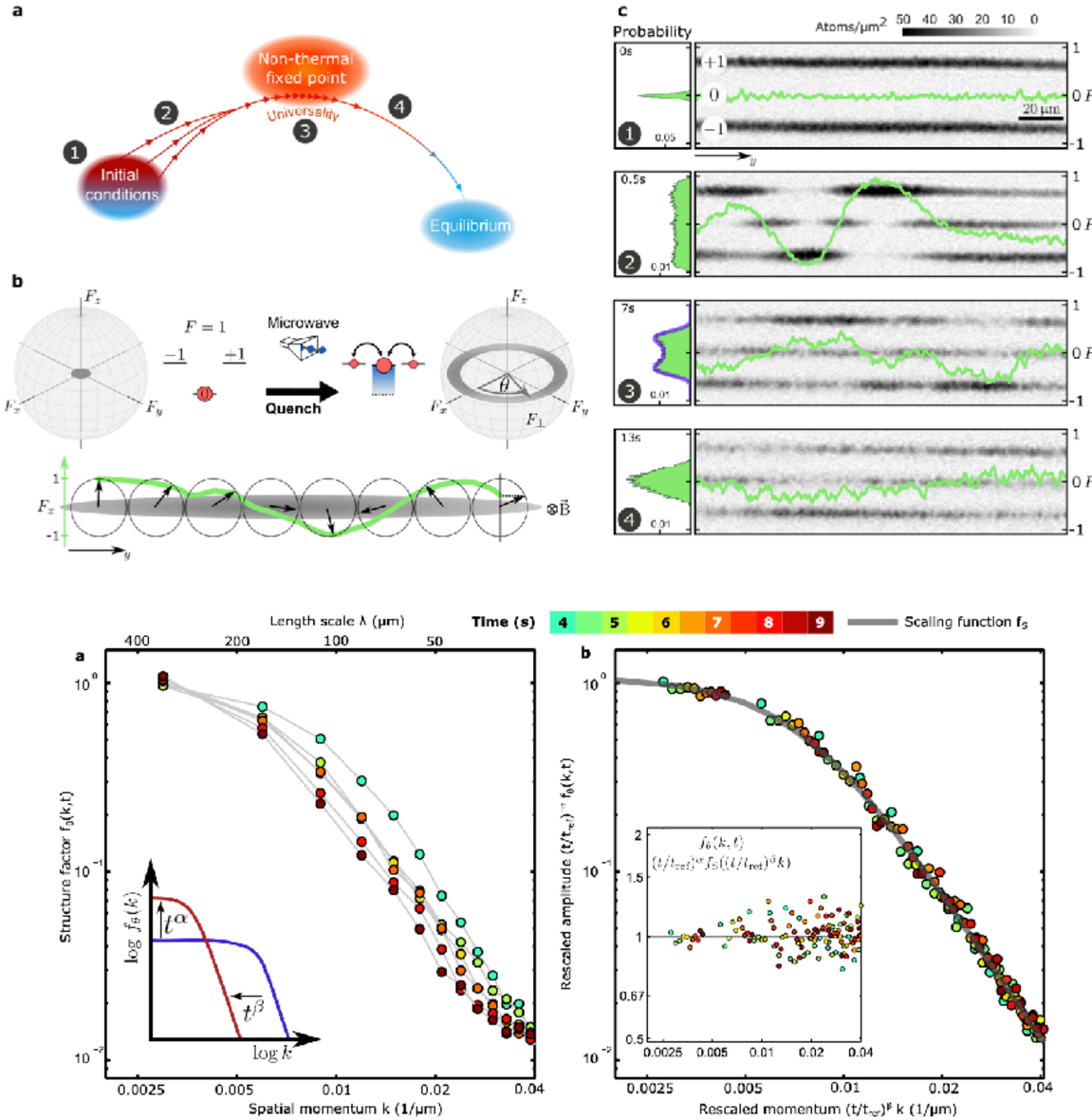
Other isolated quantum systems seem to behave similar to our QGP!

Hottest and coldest matter on earth exhibits dynamics universal to QCD at very large energies and temperatures

Another example:

3-state system in 1D trap, quenched to a non-equilibrium config.

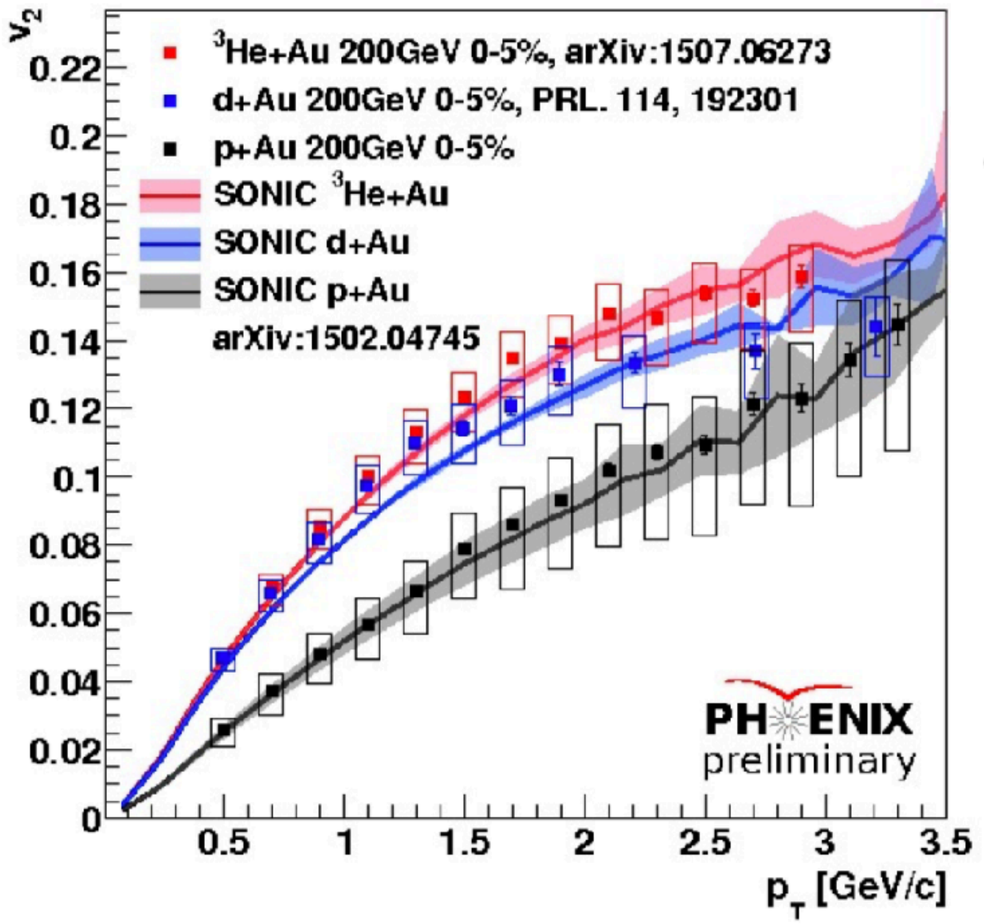
Relaxation of the system follows self-similar evolution, (with similar mechanism as in the QGP!)






Prüfer *et al.* Nature **563** (2018) 7730, 217-220

# Challenging Hydrodynamics




Smaller systems (p+p, p+A … ) exhibit similar

-Flow

-Particle suppression

-Generally, thermal fluid-like behavior

Also, we see a continuous transition between systems
Towards more flow buildup.

Heavy-Ion collisions as a laboratory to tackle the question:

**What is a fluid?**

# Summary



- To understand hot nuclear matter, we create a drop of the tiniest, hottest fluid in the Universe here, on Earth.
  - Quantum dynamics give rise to classical, fluid-like behavior
    - Fluid behaviour → Collective, correlated properties
      - We use the phenomenologically available observables to extract HotQCD quantities

# Outlook

- Using smaller systems, we challenge the current paradigms of fluids
- Non-equilibrium dynamics
  - How does nuclear matter thermalize?
- Angular momentum?
  - How does it affect what we know?
- Other crazy ideas?

